CIP B106 Mobile and IoT Forensics

Case Study 3 – Echo Show 8 IoT Evidence Investigation

Digital Forensic Examination Report

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Student name	Fuseini Imoru Kantuogaa
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Examination dates (per tool output)	12–14 October 2026 (examiner workstation Kali Linux, UTC−4)

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2 Executive summary

Scope. The supplied Echo Show 8 eMMC image (7,818,182,656 bytes, GPT, 23 partitions) and its 3,758,079,488-byte userdata partition were examined on Kali Linux using loop devices, SQLite, bulk_extractor, Scalpel, foremost and PhotoRec. No cloud account or live device was used.

Key findings. The device is an Amazon model AEOAT ("athena") running Fire OS 7.5.6.0 (Android 9), built 21 September 2023, named "Amazing Assistant". It holds one local user and one Amazon account (display name "Sarfraz"). Independent sources place device activity in US Eastern Daylight Time (UTC−4). In a 49.5-hour window (21–23 October 2023) the evidence shows Prime Video titles accessed, a Photo Booth image uploaded and a video left queued, browser searches, a YouTube search task, and Amazon Music card impressions including a "money heist" track.

Limitations. No issuer checksum was available for verification; the userdata loop device was not attached strictly read-only; phone-call artefacts were not analysed in the lab record; and the records show device activity, not the identity of any person.

3 Evidence and method

3.1 Evidence register

The table lists every item visible in the examination record. Paths are as typed on the examiner workstation.

#	Item / path	Size (bytes)	Handling	Result / note
E1	~/disk_images/EMMC_8GTF4R_ROM1_000000000000_0001D2000000.7z	2,396,000,768	7-Zip 26.00 x; LZMA2:24, 1 block	"Everything is Ok" (archive CRC). *Exhibit 1*
E2	EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin (extracted)	7,818,182,656	md5sum; loop device; carving input	MD5 b58fa04967161c158723c7b00a636533. *Exhibit 2*
E3	~/disk_images/echoshow8_extracted/…0001D2000000.bin	7,818,182,656 (same name)	Second copy used for loop0, foremost	Hash not shown; assumed identical to E2 (not verified).
E4	userdata_partition.bin (derived)	3,758,079,488	dd bs=512 skip=7929856 count=7339999	Matches mmls slot 022 length 7,339,999 sectors. *Exhibit 32* Hash not recorded.
E5	readme.txt	30	Present in working folder	Content not captured in the record.
E6	Derived outputs: bulkext/, recovered-xml/, rec_jpg_foremost/, rec_mp4_foremost/, rec_elf_photorec/, avod_recovered/, /tmp working DB copies	–	Tool output / copies	Analytical copies; not evidence originals.

3.2 Integrity check

Only one hash was recorded (MD5 of the extracted image, E2). The brief states that a checksum should be verified "where provided"; none is visible in the record, so the result is **not verified against an issuer value**. The 7z archive passed its internal integrity check, and the image size (7,818,182,656) equals the mmls end sector plus one (15,269,888 sectors × 512), which supports internal consistency but not authenticity. No post-examination re-hash is recorded.

3.3 Tools and parsers

Tool	Version (as shown)	Use
7-Zip	26.00 (x64)	Archive extraction
losetup / lsblk / mount / df	Version not captured	Loop devices, partition scan, mounting
mmls (Sleuth Kit)	Version not captured	GPT partition layout
sqlite3	3.46.1	Querying accounts, History, Prime Video, Photos databases
bulk_extractor	2.1.1	Feature extraction (email, url, exif, telephone, etc.)
Scalpel	1.60	JPEG carving of userdata partition
foremost	1.5.7	JPEG and MP4 carving of the full image
PhotoRec	7.2 (Feb 2024)	Carving of fragments (ELF-labelled output)
exiftool	13.50	Metadata of carved JPEGs

Tool	Version (as shown)	Use
ImageMagick display, jq, strings, grep, date	Version not captured	Viewing, JSON, string search, epoch conversion

3.4 Mount method and handling deviations

The eMMC image was first attached read-only with partition scanning (/dev/loop1, RO = 1) (*Exhibit 3*). A second attachment (/dev/loop0) was then created with losetup -fP **without the read-only flag** (*Exhibit 6*), and lsblk shows RO = 0 for it (*Exhibit 7*). The system partition (loop0p21) was mounted with -o ro (*Exhibit 8*); the mount options for the later userdata mount (loop0p23 at /mnt/echoshow8, *Exhibit 35*) and for a second mount at /mnt/echoshow8-2 are **not captured**. Work was performed as root, and some live SQLite databases with write-ahead logs (accounts_ce.db, accounts_de.db, History) were opened in place, whereas others (map_data_storage_v2.db, dbplaybackhistory, mixtape.db) were copied to /tmp first. These points are recorded as handling limitations: a re-hash of the image after examination, and a read-only mount for all partitions, would remove the doubt. No evidence of modification of the image is visible, but none is excluded either.

3.5 Timestamp conversion method

Formats were identified from each source: Unix epoch **milliseconds** (users, usagestats names, Prime Video, Photos, accounts); Unix epoch seconds (Bluetooth device record, build.date.utc); Chromium/WebKit **microseconds since 1601-01-01 UTC** (History); ISO-8601 with explicit −0400 offset (alexa-anchor logs); and local textual times (accounts debug_table, photo file names). Conversions used date -u (after dividing milliseconds by 1,000) and an equivalent calculation. All times are stated in **UTC and in EDT (UTC−4)**. EDT is supported by the −0400 offsets in the alexa-anchor logs, the ucid value containing America/New_York, and the photo file-name/UTC comparison in Section 4.5. The examiner workstation also reported −04:00 in exiftool, so extraction times are in the same zone but are not evidence times.

4 Device and user profile

4.1 Image scope and partition structure

mmls reports a GUID Partition Table with 512-byte sectors and 23 partitions (*Exhibit 4, 5*). Named partitions include kb, dkb, tee1/tee2, persist, misc, recovery, swdl, metadata, boot_para, nvcfg, vendor, lk/lk2, boot, **system** (slot 020, 6,356,992 sectors ≈ 3.0 GiB), **cache** (slot 021, 524,288 sectors) and **userdata** (slot 022, sectors 7,929,856–15,269,854, 7,339,999 sectors ≈ 3.5 GiB). lsblk reports a 7.3 GiB device with matching partition sizes (*Exhibit 3*). The userdata partition image (E4) was created from the full image, so its relationship to E2 is proven by offset and length, not merely assumed. File-system types were not displayed in the record; the mounted partitions were readable as Linux/Android file systems.

4.2 Device identification

Attribute	Value	Source
Brand / model / device	Amazon / AEOAT / athena	/system/build.prop (ro.product.*) *Exhibit 10*
Operating system	Fire OS 7.5.6.0 (PS7560/3677), Android 9, SDK 28	build.prop *Exhibit 10, 11*
Build date	Thu Sep 21 02:41:50 UTC 2023 (ro.build.date.utc 1695264110)	build.prop *Exhibit 10*
Security patch level	2022-01-01	build.prop *Exhibit 10*
Build number	0028421545348	build.prop *Exhibit 9, 11*
Hardware context	32-bit ARM (armeabi-v7a), Cortex-A53; init.recovery.mt8183.rc present	build.prop *Exhibit 11*; root listing *Exhibit 8*
Device name	“Amazing Assistant”; Bluetooth name “Echo Show 8 (2nd Gen)-3SD”	deviceNameSharedPref.xml *Exhibit 13*; bt_config.conf

Attribute	Value	Source
Serial / device ID	G6G1…03SD (masked)	device_capabilities_api.amazonalexa.com (cache file); matches “3SD” in Bluetooth name

The slides describe an Echo Show 8; the image evidence agrees (“Echo Show 8 (2nd Gen)”, model code AEOAT). The photo database also records image model AEOAT (*Exhibit 44*), so model identity is supported by three independent places. No discrepancy needing a limitation was found, although the Bluetooth name is a user-adjustable label and the model code is the firmer value.

4.3 Acquisition context and provenance

The course material describes teardown, chip-off and imaging of the eMMC. The supplied file name (EMMC_8GTF4R_ROM1…) and the full-size 7.3 GiB raw image fit a chip-off read of the eMMC user area. That supports the **provenance of the training data** only. It does not show who performed the acquisition, when, or on which physical unit, and nothing in the image tells us whether the device was powered, locked or in use at acquisition. It cannot show what the device’s cloud account holds, and it says nothing about any person’s conduct.

4.4 Users, accounts and application context

- **Local users.** userlist.xml and sorteduserlist.xml list a single user, id 0; guest restrictions disable SMS, outgoing calls, Wi-Fi configuration and unknown-source installation (*Exhibit 14*). 0.xml names user 0 “Sarfraz”, with lastLoggedIn **1698016596033** = 2023-10-22 23:16:36 UTC (19:16:36 EDT) (*Exhibit 15*). Its stored build fingerprint number (0028421545216) differs from the current build number (0028421545348), indicating the firmware was updated after that fingerprint was recorded.
- **Accounts.** accounts_ce.db and accounts_de.db contain two Amazon sync accounts, “Amazon Video Sync” and an account named “Sarfraz” (*Exhibit 18, 19*). map_data_storage_v2.db (copied to /tmp) lists a single Amazon account with the same display name. The Prime Video profile cache lists an adult profile “Sarfraz” (default) and a child profile; identifiers are masked here.

- **Account history.** debug_table shows account additions on 2021-02-18 03:11:18, 2023-10-02 21:00:18–21:00:32, a removal at 2023-10-09 16:34:24 and a re-addition at 16:39:41 (*Exhibit 20*).
- **Applications and skills.** /system/app contains Amazon device-metrics, WebView, Bluetooth, Music, Email, Calendar, KeyChain and a captive-portal login component (*Exhibit 12*). Data directories examined include com.amazon.cloud9 (browser/WebView and shell), com.amazon.avod (Prime Video), com.amazon.zordon (Photos/Photo Booth), com.amazon.paladin (alexa-anchor logs), com.amazon.kindle.otter.oobe (setup cache) and amazon.speech.sim.
- **Usage statistics.** Nine daily usagestats files exist (*Exhibit 16*); the content of the 22 October file shows counts of interactive and non-interactive periods and per-package activity records (*Exhibit 17*).

4.5 Cross-source relationships

C1 – Time zone and account timing. accounts_de.db stores the "Sarfraz" account creation as epoch 1696883981793 (2023-10-09 20:39:41 UTC), while debug_table records the same add as 2023-10-09 **16:39:41**: the two differ by exactly four hours, so debug_table uses device-local EDT on that date. A cached Alexa address-suggestions response in the setup cache carries a server date of 20:40:08 GMT (X-Android-Received-Millis 1696884008767), 27 seconds after the account add – a plausible setup sequence. The street address in that response is masked in this report; it is a retail default shipping address and not evidence of device location.

C2 – Photo timestamps. For the Photo Booth image taken on 21 October, the file name reads 2023-10-21_19-02-39, the cloud node was created at epoch 1697929372480 (23:02:52 UTC = 19:02:52 EDT) and the EXIF-style date field is 1697914959000, which converts to 19:02:39 if read as UTC – i.e. local time stored as if it were UTC. The three values agree with EDT and with each other (*Exhibit 43, 44, 45*).

C3 – First setup. bt_config.conf was created on 2021-02-18 03:10:55 and the first account add is 03:11:18, 23 seconds later. accounts_de.db stores the same event as epoch 1613617878925 = 2021-02-18 03:11:18 **UTC**, equal to the debug_table text. On 2 October 2023 the debug_table times (21:00:18–21:00:32) again equal UTC, yet photo file names that evening are in EDT. The device clock zone therefore appears to have been UTC at set-up and EDT afterwards (changed between 2 and 9 October). This is an inference from timestamp patterns, not a recorded settings change.

4.6 Missing, overwritten or deleted data

Carving recovered earlier copies of the alexa-anchor log (Section 5.6), which shows that older log content survives in unallocated or reused space. Several expected items were not examined or were not shown: persist partition contents, system logs, deleted SQLite records (the History journal files exist but were not parsed), and the readme.txt. Android maintenance, log rotation and flash wear-levelling are sufficient explanations for missing or partial records; no deliberate removal is implied.

5 Artefact findings

5.1 Network connectivity

Wi-Fi. /misc/wifi/WifiConfigStore.xml lists two saved WPA-PSK networks (SSIDs partially masked: Fios-pJ… and TellMy…). The record shows Priority 0, null BSSID, Status 2 (enabled), and for one entry ValidatedInternetAccess true, NoInternetAccessExpected false. These are **configuration records**: they show networks the device is set up to use, not a time-stamped connection event, and the pre-shared key stored in the file is deliberately not reproduced. The null BSSID means no access-point hardware address was stored. device_disconnection_times.xml and com.amazon.settings.wifi.xml exist in the settings shared preferences (*Exhibit 13*) but were not opened in the record.

Bluetooth. bt_config.conf names the adapter "Echo Show 8 (2nd Gen)-3SD" and two paired remote devices: a laptop named after the account holder (MAC F4:5C:89:xx:xx:xx masked) and a portable speaker "onn. Rugged Speaker Mini" (record timestamp 1696864570 = 2023-10-09 15:16:10 UTC, 11:16:10 EDT, interpreted as the bonding time but not confirmed). Bt_Metrics_Device_Connected.xml marks the laptop as connected at the time of the last metrics write. Snapshot 623 shows the on-device Bluetooth screen naming the same laptop (*Exhibit 24*). Pairing data identifies devices, not their users.

Corroboration. The laptop MAC appears in both bt_config.conf and Bt_Metrics_*.xml, and the same device name is on screen in snapshot 623: three independent sources agree on the pairing. An alternative explanation for the presence of the laptop name is that the pairing was made by anyone with access to the laptop during set-up.

5.2 Web activity

The Chromium-based WebView profile of com.amazon.cloud9 contains History, Cookies, Web Data, Bookmarks and caches (*Exhibit 28*). The History visits table (*Exhibit 29*) and urls table (*Exhibit 31*) hold 126 visits to 88+ URLs. Chromium timestamps are WebKit microseconds; for example last_visit_time 13342492250476128 = 2023-10-22 23:50:50 UTC (19:50:50 EDT).

- **Searches.** keyword_search_terms records “search chicken nuggets on google” and “close google” (*Exhibit 30*); the urls table dates them 19:50:50 EDT and 19:50:12 EDT on 22 October. The terms are typed or voice-transcribed text in a search box; the artefact does not distinguish the two.
- **Streaming and sign-in pages.** YouTube searches and Shorts pages are dated 9 October 18:30–18:36 EDT (e.g. last_visit 13341364600297498 = 22:36:40 UTC); on 21 October 20:17–20:29 EDT the profile visited Microsoft sign-in pages whose redirect target is alexa.amazon.com with calendar and contact permissions (an account-linking flow; the account identifier in the URL is masked). A Bing page was visited at 19:46:41 EDT and Microsoft account pages at 19:51:08 and 19:52:55 EDT on 22 October.
- **Recent tasks.** 617_task.xml records an android.intent.action.SEND for a YouTube search for “kapil sharma show” launched from com.amazon.cloud9 (SlateEchoActivity), last_time_moved 1698022572750 = 2023-10-23 00:56:12 UTC (20:56:12 EDT, 22 October) (*Exhibit 25*). The matching snapshot 617.jpg shows a video page on the device (*Exhibit 22*), and snapshot 627 shows an on-screen “Exercise” reminder at 7:00 PM (*Exhibit 23*).

Corroboration. The task XML (task id 617), the snapshot file 617.jpg and the browser/shell package (com.amazon.cloud9) agree on the YouTube search. The History urls table also holds a similar “Kapil sharma” search dated earlier (13342240539770331 = 2023-10-20 01:55:39 UTC); the task therefore may reflect a repeat or a continued session. These artefacts show a request and a rendered page on the device, not who made the request or whether the video was watched through.

5.3 Phone communication

Result: not established from the examination record. The lab record contains no call-log, contact or messaging database. Sources that exist but were not interpreted: Bluetooth permission preferences (phonebook_access_permission.xml, message_access_permission.xml and

sim_access_permission.xml in com.android.bluetooth shared_prefs); the bulk_extractor telephone.txt and vcard.txt feature files (*Exhibit 27*); and the guest restrictions in userlist.xml (no_sms and no_outgoing_calls true, *Exhibit 14*). A Bluetooth hands-free profile (HFP audio-gateway) UUID is recorded for the paired laptop, which shows the capability, not any call. No communication event is reported. This absence reflects the sources reviewed; it does not prove that no calls occurred.

5.4 Photos

- **Photo Booth uploads.** com.amazon.zordon upload_requests lists seven JPEGs from 2 October to 21 October 2023 and two videos (*Exhibit 42, 43*). Each completed row has a file name containing local time, a size and an MD5 (e.g. the 21 October image: 2,961,976 bytes, MD5 f2acb3ee…). cloud_nodes metadata gives image make Amazon and model AEOAT, 2160-pixel height and a "custom software" tag (*Exhibit 44*). This shows the pictures originated from the device's own camera application, not who was in the frame.
- **Recovered images.** Scalpel's workload on the userdata partition was 12 + 6,296 JPEG header matches (*Exhibit 33, 34*) and foremost 6,390 in the full image (*Exhibit 37*). The counts differ because the tools use different header signatures and scope (Appendix E). Most carved items are 480×480 artwork thumbnails for video and music titles (*Exhibit 38, 39*). Two recovered userdata images show a seated adult indoors and a family group at a tourist attraction. They are not reproduced here; the persons are not identified, and there is no metadata in the record linking them to a named account.
- **Snapshots.** Five task snapshots exist, with "reduced" copies (*Exhibit 21*). A find on the mounted userdata shows only snapshots and two Alexa skills-image files as JPEGs in the live file system.

5.5 Videos

Foremost found no MP4 files in the full image (*Exhibit 46*). Video evidence is therefore taken from databases: upload_requests row 4 is a Photo Booth video file (photo_booth_video1696880081230.mp4, 6,047,662 bytes, DONE) whose name converts to 2023-10-09 19:34:41 UTC; row 8 (photo_booth_video1697929429134.mp4, 13,958,448 bytes, name converts to **2023-10-21 23:03:49 UTC / 19:03:49 EDT**) is in state **QUEUED** with no node id, no bytes sent and no MD5 (*Exhibit 43*). A queued item may exist only as a pending upload and may not be present on the device. Duration,

thumbnail and sidecar files were not reviewed; no playback-derived times are used. The Prime Video database (*Exhibit 40, 41*) records five titles with lastAccessed times (Section 6) and a watched position and runtime for each, e.g. one film at about 1 h 42 min of a 2 h 10 min runtime. This is a viewing-history record, not a media file.

5.6 Audio and Alexa interface logs

No audio recording files were examined. The com.amazon.paladin/alexa-anchor log (file.0000…237 with a parallel message-timestamps file, *Exhibit 48*) contains DeviceImpression events with ISO timestamps at −0400. One event, at 2023-10-23T18:28:44.594−0400 (22:28:44 UTC), is a RecentlyPlayedAudioContentProvider card whose ucid names Amazon Music, media type SONG, “money heist theme song (Bella Ciao – lofi edit)” (*Exhibit 47*). Every event has the device state camera “ON”, cameraShutter “OFF” and privacyMode “OFF”, which describes the device configuration at display time only.

PhotoRec’s carved fragments contain 14 files matching “money” (*Exhibit 49, 50*). One (f5777032.elf) holds an earlier version of the same log, with an impression for the same song at **2023-10-22T22:12:50−0400** (02:12:50 UTC on 23 October) (*Exhibit 51*). A further carved fragment holds timed-text subtitle lines, which are consistent with a feature film’s dialogue; content is not reproduced. **Corroboration:** the same title appears in a live log entry and in a carved earlier copy, on different days. **Qualification:** a RecentlyPlayed card shows the item was on screen as a recent-item suggestion; it is not an exact play time, and no voice request, transcript or playback event is present.

6 Event chronology

Window justification: 2023-10-21 18:00 EDT to 2023-10-23 19:30 EDT (22:00 UTC on 21 October to 23:30 UTC on 23 October), **49.5 hours**. It covers the earliest dated item that clearly belongs to the planted activity (the Photo Booth capture), the daily-usage cluster of 22 October, and the last alexa-anchor events. The window contains 16 entries from six artefact families (Photos, Prime Video, browser history, system/user, recent tasks and usagestats, Alexa

interface logs) and five corroborated relationships (C1–C5 in Sections 4.5 and 5, and the table below). Earlier events (e.g. 2–9 October account and photo activity) are in the finding log only.

#	Time EDT (UTC−4)	Time UTC	Event	Source / raw value	Qualification
T1	2023-10-21 18:24:48	2023-10-21 22:24:48	Browser visit to an Amazon “Alexa and Alexa Device FAQs” page	History urls id 52; 13342400688326954 (WebKit µs)	Page visited; a visit, not proof of reading.
T2	2023-10-21 19:02:52	2023-10-21 23:02:52	Photo Booth image created and uploaded (DONE)	zordon mixtape.db cloud_nodes 7; 1697929372480 ms. File name …19-02-39-687; EXIF-style 1697914959000	C2 – name, node time and image time agree. Device camera application; subject not identified.
T3	2023-10-21 19:03:49	2023-10-21 23:03:49	Photo Booth video file name created; upload remains QUEUED	upload_requests row 8; name epoch 1697929429134	Name time only; no node, MD5 or bytes sent. May not be on device.
T4	2023-10-21 19:46:51	2023-10-21 23:46:51	Prime Video: “Jurassic World Dominion” lastAccessed	avod dbplaybackhistory; 1697932011023 ms	Viewing-history field; not a start or end time.

#	Time EDT (UTC−4)	Time UTC	Event	Source / raw value	Qualification
T5	2023-10-21 20:17:12–20:29:19	2023-10-22 00:17:12–00:29:19	Microsoft sign-in pages with redirect to Alexa account-linking	History urls 53–57; 13342407432028665 to 13342408159065431	Linking flow requested; completion not shown. Identifier masked.
T6	2023-10-21 20:45:12	2023-10-22 00:45:12	Daily usagestats file written	usagestats/0/daily/1697935512409	File name epoch; 11 interactive periods in that file. Day-boundary file.
T7	2023-10-22 19:09:09	2023-10-22 23:09:09	Prime Video: “Top Gun: Maverick” lastAccessed	dbplaybackhistory; 1698016149065 ms	Position about 1 h 42 m of 2 h 10 m recorded; viewer not identified.
T8	2023-10-22 19:16:36	2023-10-22 23:16:36	Local user 0 lastLoggedIn updated	system/users/0.xml; 1698016596033 ms	Manually re-converted (Appendix E). Setting record; not a person.
T9	2023-10-22 19:46:41	2023-10-22 23:46:41	Bing home page visited	History urls 65–66; 13342492001629428	Page visit.
T10	2023-10-22 19:50:12–19:52:55	2023-10-22 23:50:12–23:52:55	Google searches “close google” and “search chicken nuggets on google”; Microsoft account pages	History urls 69–73; keyword_search_terms	Search terms recorded; typed vs spoken not distinguishable.

#	Time EDT (UTC−4)	Time UTC	Event	Source / raw value	Qualification
T11	2023-10-22 20:43:53	2023-10-23 00:43:53	Last recorded browser visit	History visits id 126; 13342495433605931	Last row in visits table.
T12	2023-10-22 20:52:12	2023-10-23 00:52:12	Latest daily usagestats file	usagestats/0/daily/1698022332424	File-name epoch.
T13	2023-10-22 20:56:12	2023-10-23 00:56:12	Shell task launches YouTube search “kapil sharma show”	recent_tasks/617_task.xml; 1698022572750 ms; snapshot 617.jpg	C4 – task XML, snapshot and package agree. Request and rendered page; user unknown.
T14	2023-10-22 22:12:50	2023-10-23 02:12:50	Carved alexa-anchor impression: Amazon Music SONG “money heist theme song…”	PhotoRec f5777032.elf; “2023-10-22T22:12:50.917−0400”	C5 – same title in live log at T16. Carved fragment; card not a play event.
T15	2023-10-23 17:11:46; 17:14:05; 17:28:48	2023-10-23 21:11:46; 21:14:05; 21:28:48	Prime Video: “Sicario”, “The Peripheral” (Pilot), “Memory” lastAccessed	dbplaybackhistory; 1698095506321; 1698095645376; 1698096528354	Three titles within 17 minutes; each is a last-access value.

#	Time EDT (UTC−4)	Time UTC	Event	Source / raw value	Qualification
T16	2023-10-23 18:28:05–19:21:44	2023-10-23 22:28:05–23:21:44	alexa-anchor impressions; 18:28:44 RecentlyPlayed “money heist theme song”	paladin file.…237 and message-timestamps; ISO −0400	C5 – corroborates T14. Timestamps carry explicit zone.

Corroboration key: **C1** account add vs debug_table (4 h offset); **C2** photo file name vs node time vs image time; **C4** task XML vs snapshot vs package; **C5** “money heist” title in live and carved logs; **C6** model AEOAT in build.prop, Bluetooth name and photo metadata.

7 Interpretation and limits

7.1 Four findings

Finding 1 – Device and configuration context. The image is an Echo Show 8 (model AEOAT, “athena”) running Fire OS 7.5.6.0 on Android 9, built on 2023-09-21 UTC, named “Amazing Assistant”, with one local user and one Amazon account. The model is supported by build.prop, the Bluetooth adapter name and photo metadata. *Limits:* a name or model does not identify an owner; the firmware appears to have been updated after the last recorded login fingerprint.

Finding 2 – User and application activity. The account, profile and user records consistently use the display name “Sarfraz”. Prime Video history shows five titles accessed between 21 and 23 October 2023, and the photo uploader shows device-camera images on 2, 7, 9, 17 and 21 October. *Alternative explanations:* a shared household device, a guest, or a family profile (a child profile also exists); a display name proves an account label, not a person.

Finding 3 – Network, web, communication and media activity. Two Wi-Fi networks and two Bluetooth peers are configured; the browser shows YouTube, Bing, Google and Microsoft sign-in page visits including two searches on 22 October; Alexa interface logs show an Amazon Music "money heist" card; one Photo Booth video was left queued; and no phone-call evidence is reported. *Limits:* configuration is not connection; a card display is not an audio play; no media content was transcribed or compared.

Finding 4 – Reconstructed chronology. Across six artefact families, device activity clusters on the evenings of 21, 22 and 23 October 2023 (EDT), with browser, usage and task artefacts concentrated between 19:09 and 20:56 EDT on 22 October. Time handling is the greatest source of risk: seven formats were used, and the device clock zone appears to have changed between 2 and 9 October. *Limits:* "lastAccessed" and "last_visit_time" are last values, so earlier events are overwritten; the order of events across sources assumes the device clock was accurate, which was not tested against an external time source.

7.2 What remains uncertain

- Authenticity and completeness of the supplied image (no issuer hash).
- Whether the loop0 attachment or live database opens changed any evidence data (no re-hash).
- Phone communication, voice command history and cloud-held data (not present or not examined).
- Whether searches were typed or spoken, and who was present at the device.
- The precise meaning of a few fields (usagestats lastTimeActive negative values; Bluetooth device timestamp).

7.3 Contradictions

The date command in *Exhibit 15* printed year 55778 because the millisecond value was entered as seconds; this is an examiner error, not a data conflict, and is corrected in Appendix E. The scalpel and foremost carve counts differ (6,308 header matches vs 6,390 files); this reflects different signature sets and is not a contradiction about the image. No two artefacts conflict on time zone or identity once the 2021/2023 UTC-to-EDT difference in debug_table is understood.

Appendix A – Finding log

IDs F01–F23. Times are given in UTC and EDT (UTC−4) unless stated. Identifiers are masked.

ID	Finding	Source	Raw time or value	Converted time	Qualification
F01	Image is a 7,818,182,656-byte GPT eMMC image with 23 partitions; userdata is slot 022 (7,339,999 sectors)	mmls on EMMC_8GTF4R…0001D2000000.bin	15,269,888 sectors	Image size 7.3 GiB	Internally consistent; no issuer hash to verify.
F02	MD5 of extracted image recorded	md5sum of extracted .bin	b58fa04967161c158723c7b00a636533	–	Not compared with an issuer value.

ID	Finding	Source	Raw time or value	Converted time	Qualification
F03	Device model AEOAT, device “athena”, Fire OS 7.5.6.0 (Android 9, SDK 28)	system partition /system/build.prop	ro.product.model=AEOAT; ro.build.version.name	–	Corroborated by photo metadata (AEOAT) and Bluetooth name.
F04	Firmware build date	build.prop ro.build.date.utc	1695264110	2023-09-21 02:41:50 UTC (2023-09-20 22:41:50 EDT)	Matches device’s own textual build date.
F05	Device name “Amazing Assistant”; serial masked G6G1…03SD	settings deviceNameSharedPref.xml; amazon.speech.sim capabilities file	deviceNameKey	–	Name is user-editable; serial matches “3SD” in Bluetooth name.

ID	Finding	Source	Raw time or value	Converted time	Qualification
F06	Single local user id 0 “Sarfraz”; guest restrictions set	/data/system/users/0.xml, userlist.xml	lastLoggedIn=1698016596033	2023-10-22 23:16:36 UTC (19:16:36 EDT)	Setting record; not proof of a person. Fingerprint build differs from current build.
F07	Amazon accounts: two sync accounts, “Amazon Video Sync”, “Sarfraz”	accounts_ce.db / accounts_de.db; map_data_storage_v2.db	creation 1696883981793	2023-10-09 20:39:41 UTC (16:39:41 EDT)	C1: equals debug_table 16:39:41.
F08	First account add on device	accounts_de.db row 1; debug_table row 1	1613617878925	2021-02-18 03:11:18 UTC	Equals debug_table text; bt_config.conf created 03:10:55.

ID	Finding	Source	Raw time or value	Converted time	Qualification
F09	Account removed and re-added on 9 Oct 2023	accounts_de.db debug_table	textual	2023-10-09 16:34:24 / 16:39:41 (EDT)	Local device time; precise reason unknown.
F10	Two saved Wi-Fi networks (WPA-PSK), priority 0, null BSSID	/misc/wifi/WifiConfigStore.xml	Status=2; ValidatedInternetAccess=true (one)	No timestamp	Configuration only; key not reproduced; SSIDs masked.
F11	Bluetooth adapter “Echo Show 8 (2nd Gen)-3SD”; paired laptop and “onn. Rugged Speaker Mini”	/misc/bluedroid/bt_config.conf; com.android.bluetooth shared_prefs	Timestamp=1696864570 (speaker)	2023-10-09 15:16:10 UTC (11:16:10 EDT)	Pairing time inferred; keys and MACs not reproduced.

ID	Finding	Source	Raw time or value	Converted time	Qualification
F12	Browser searches “search chicken nuggets on google” and “close google”	com.amazon.cloud9 amazon_webview/History: keyword_search_terms, urls	13342492250476128; 13342492212049982	2023-10-22 23:50:50 / 23:50:12 UTC (19:50:50 / 19:50:12 EDT)	Typed/voice indistinguishable.
F13	YouTube search task “kapil sharma show” by shell activity	/data/system_ce/0/recent_tasks/617_task.xml	last_time_moved=1698022572750	2023-10-23 00:56:12 UTC (20:56:12 EDT, 22 Oct)	C4: snapshot 617.jpg; requester unknown.

ID	Finding	Source	Raw time or value	Converted time	Qualification
F14	Prime Video "Top Gun: Maverick" last accessed	com.amazon.avod/files/databases/dbplaybackhistory	1698016149065	2023-10-22 23:09:09 UTC (19:09:09 EDT)	Last-access value.
F15	Prime Video "Jurassic World Dominion" last accessed	dbplaybackhistory	1697932011023	2023-10-21 23:46:51 UTC (19:46:51 EDT)	As F14.

ID	Finding	Source	Raw time or value	Converted time	Qualification
F16	Prime Video “Sicario”, “The Peripheral” (Pilot), “Memory” last accessed	dbplaybackhistory	1698095506321 / 1698095645376 / 1698096528354	2023-10-23 21:11:46 / 21:14:05 / 21:28:48 UTC (17:11–17:28 EDT)	As F14.
F17	Photo Booth image uploaded 21 Oct 2023	zordon mixtape.db cloud_nodes id 7; upload_requests id 7	1697929372480; MD5 f2acb3ee…	2023-10-21 23:02:52 UTC (19:02:52 EDT)	C2: file name 19-02-39 and image time agree.

ID	Finding	Source	Raw time or value	Converted time	Qualification
F18	Photo Booth video file queued, not uploaded	upload_requests id 8	1697929429134; QUEUED; 13,958,448 B	2023-10-21 23:03:49 UTC (19:03:49 EDT)	No node/MD5; may not exist on device.
F19	Carved JPEGs: 6,390 (foremost); mostly 480×480 artwork	foremost audit; exiftool	n/a	Extraction time only	File times in exiftool are examiner times.
F20	No MP4 files carved from the image	foremost -t mp4 audit	0 files	–	Absence only supports “not recovered by this method”.

ID	Finding	Source	Raw time or value	Converted time	Qualification
F21	Alexa interface log: “money heist theme song (Bella Ciao – lofi edit)” RecentlyPlayed card	paladin/files/alexa-anchor/file.…237	2023-10-23T18:28:44.594-0400	2023-10-23 22:28:44 UTC (18:28:44 EDT)	C5: card, not a play time.
F22	Earlier copy of same log with same song card	PhotoRec f5777032.elf (carved)	2023-10-22T22:12:50.917-0400	2023-10-23 02:12:50 UTC (22:12:50 EDT, 22 Oct)	Carved fragment; offset in image not recorded.
F23	Phone communication artefacts	Bluetooth permission prefs; telephone.txt/vcard.txt (bulk_extractor)	–	–	Not interpreted in the record; no call event reported.

Appendix B – Labelled exhibits (screenshots)

Each exhibit is a screenshot from the examiner's own session. The caption states what it shows. Screenshots that display a street address, Wi-Fi pre-shared key, Bluetooth link keys, e-mail log-in names, personal media or long account identifiers were **withheld** from this report and are held in the case file; the findings drawn from them are stated in the text with identifiers masked.

Exhibit 1. Extraction of the supplied 7z archive (2,396,000,768 bytes, LZMA2, "Everything is Ok"; extracted size 7,818,182,656 bytes).

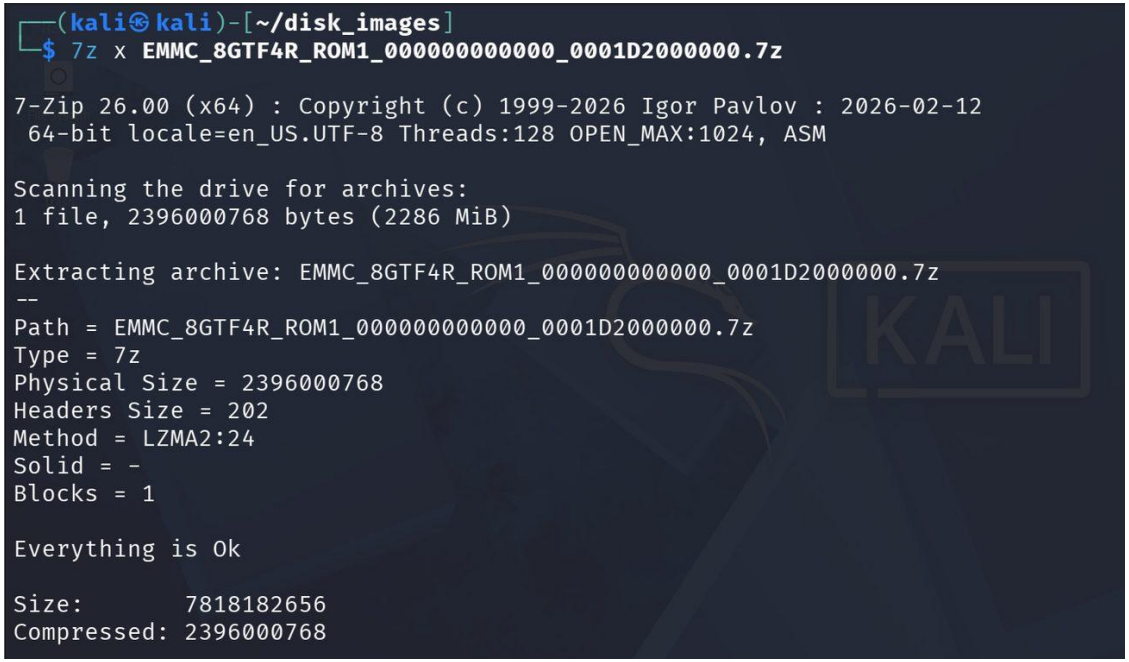


Exhibit 2. Working directory listing and MD5 of the extracted eMMC image (b58fa04967161c158723c7b00a636533). No issuer checksum is shown for comparison.



Exhibit 3. Whole-image loop device attached with --partscan --read-only (/dev/loop1); loop0 is a second attachment at offset 4060086272, size limit 3758079488 (userdata).

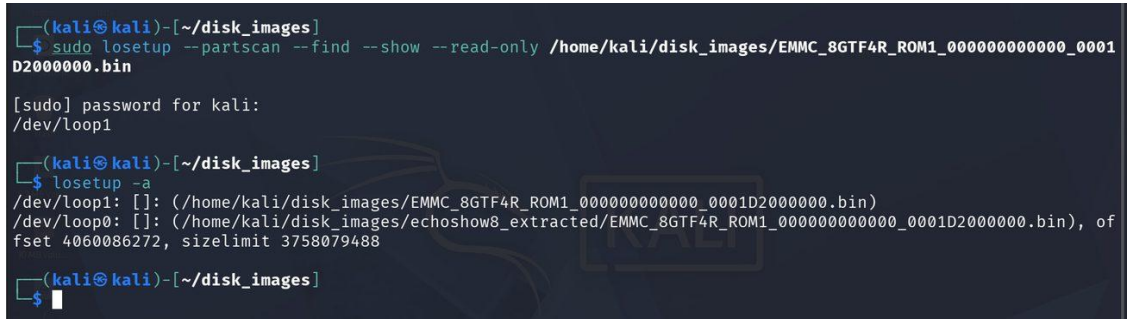
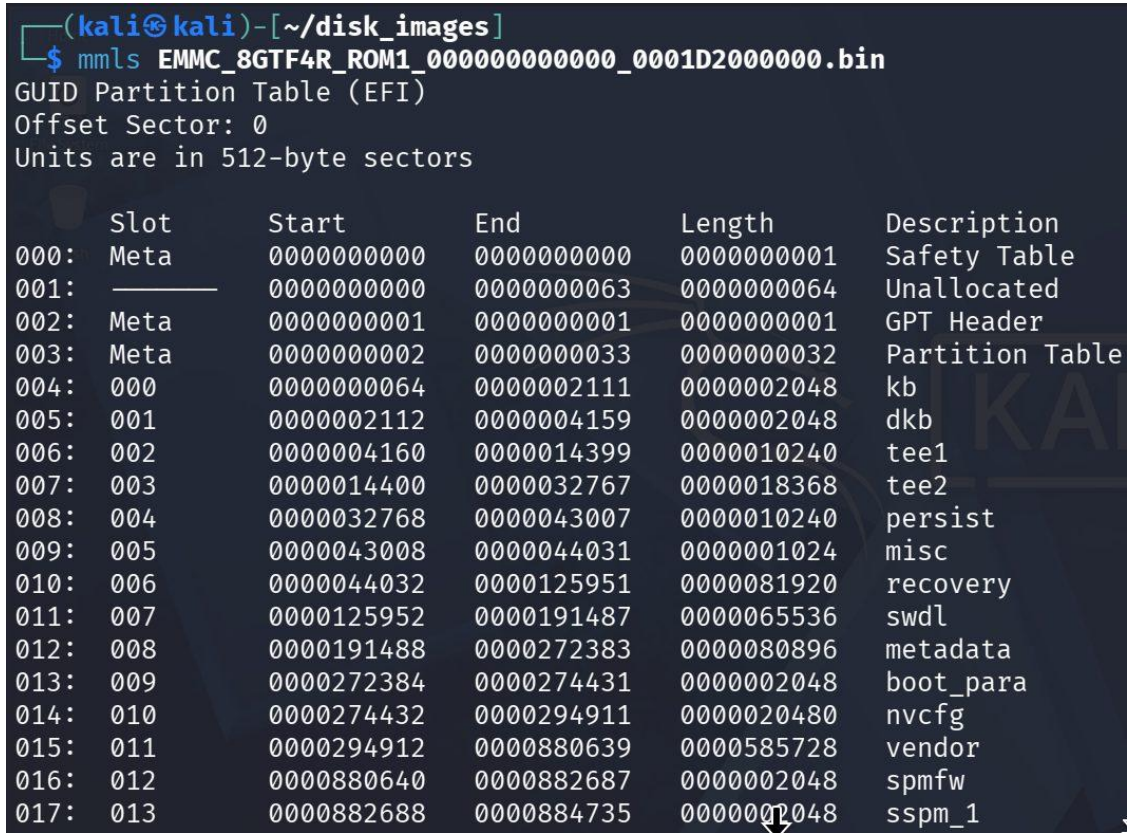


Exhibit 4. mmls (The Sleuth Kit) – GPT, 512-byte sectors, slots 000–013.



```
┌──(kali㉿kali)-[~/disk_images]
└─$ mmls EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

     Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Safety Table
001:  ———————   0000000000   0000000063   0000000064   Unallocated
002:  Meta      0000000001   0000000001   0000000001   GPT Header
003:  Meta      0000000002   0000000033   0000000032   Partition Table
004:  000       0000000064   0000002111   0000002048   kb
005:  001       0000002112   0000004159   0000002048   dkb
006:  002       0000004160   0000014399   0000010240   tee1
007:  003       0000014400   0000032767   0000018368   tee2
008:  004       0000032768   0000043007   0000010240   persist
009:  005       0000043008   0000044031   0000001024   misc
010:  006       0000044032   0000125951   0000081920   recovery
011:  007       0000125952   0000191487   0000065536   swdl
012:  008       0000191488   0000272383   0000080896   metadata
013:  009       0000272384   0000274431   0000002048   boot_para
014:  010       0000274432   0000294911   0000020480   nvcfg
015:  011       0000294912   0000880639   0000585728   vendor
016:  012       0000880640   0000882687   0000002048   spmfw
017:  013       0000882688   0000884735   0000002048   sspm_1
```

Exhibit 5. mmls – slots 014–026; system (020), cache (021) and userdata (022, sectors 7929856–15269854).

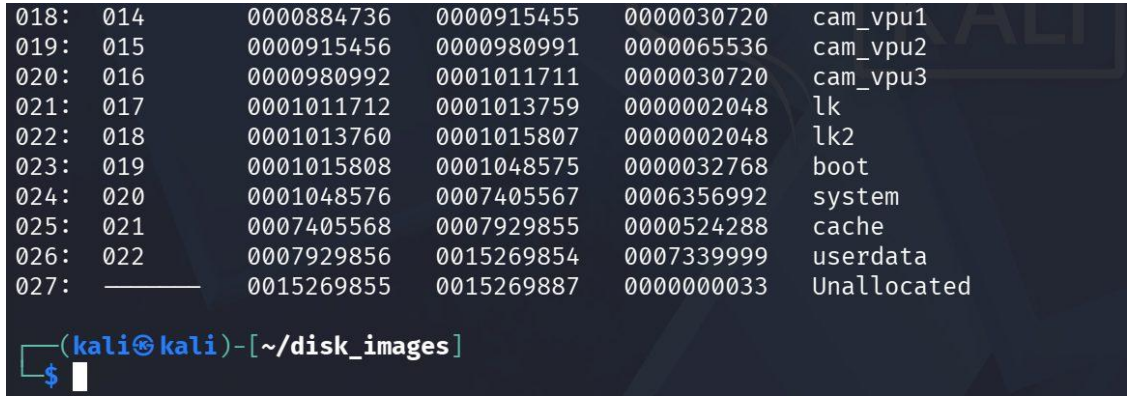


Exhibit 6. Second attachment created with losetup -fP (no read-only flag); only loop0 is listed afterwards.

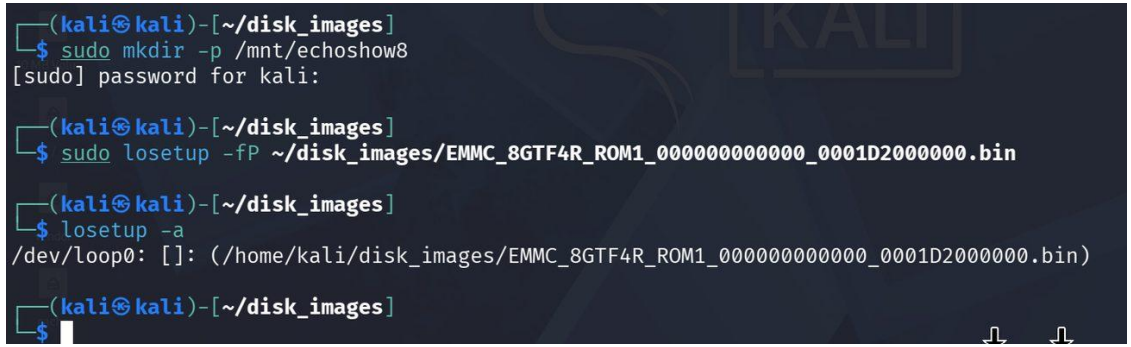


Exhibit 7. lsblk /dev/loop0 – RO column = 0; partition 23 (3.5 G) mounted at /mnt/echoshow8.

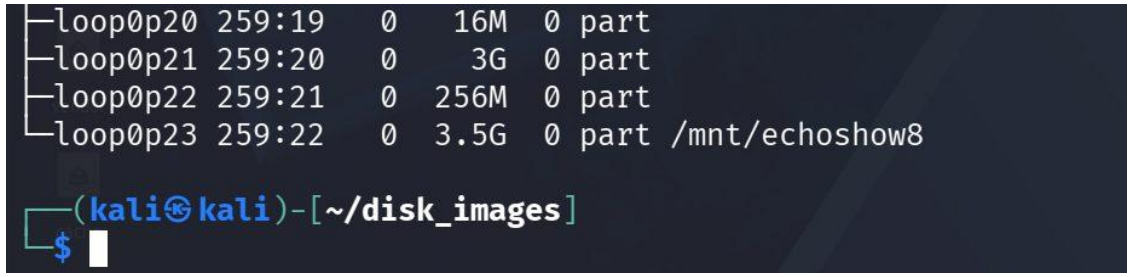


Exhibit 8. System partition (loop0p21) mounted with -o ro at /mnt/echoshow8 and its root listing (system-as-root layout).

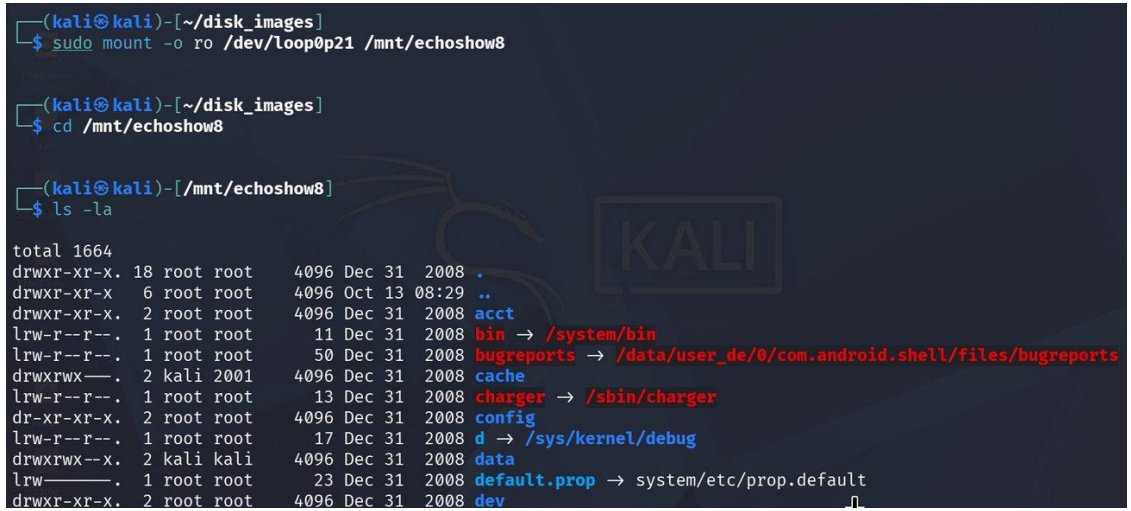


Exhibit 9. /system/build.prop (first lines): build id PS7560.3677N, incremental 0028421545348, SDK 28.

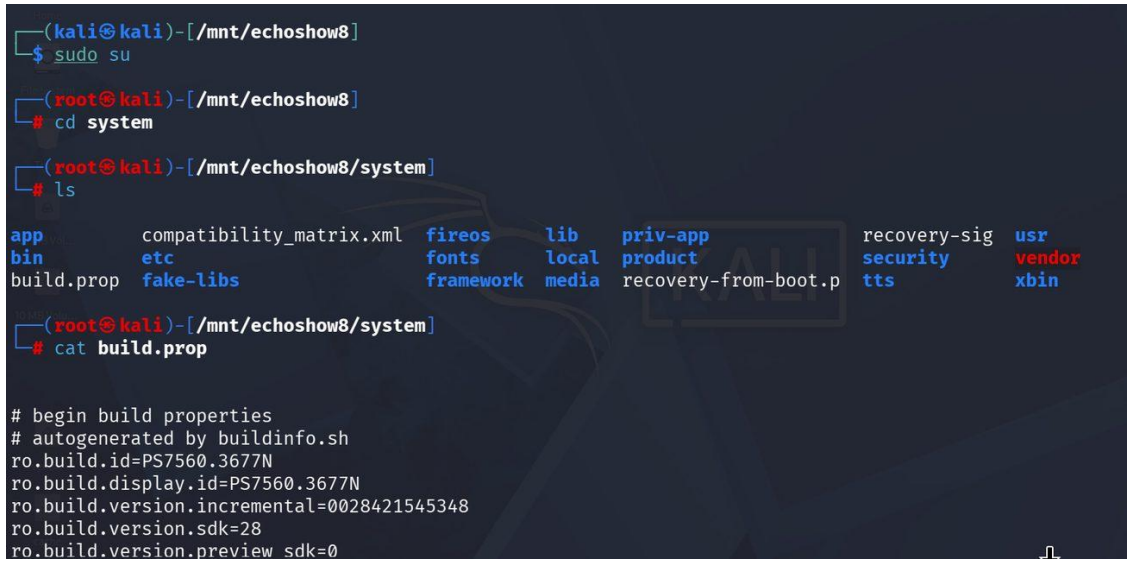
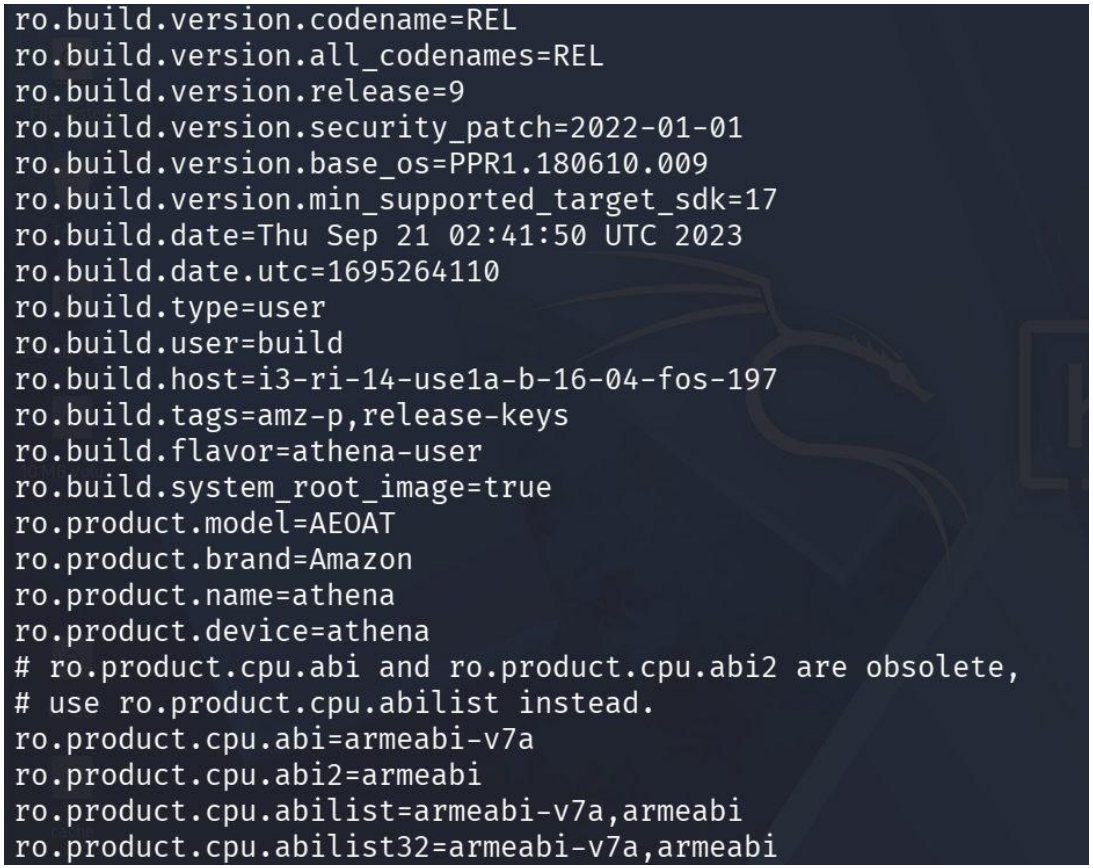


Exhibit 10. build.prop: Android release 9, security patch 2022-01-01, build date Thu Sep 21 02:41:50 UTC 2023 (1695264110), model AEOAT, brand Amazon, device “athena”.



```
ro.build.version.codename=REL
ro.build.version.all_codenames=REL
ro.build.version.release=9
ro.build.version.security_patch=2022-01-01
ro.build.version.base_os=PPR1.180610.009
ro.build.version.min_supported_target_sdk=17
ro.build.date=Thu Sep 21 02:41:50 UTC 2023
ro.build.date.utc=1695264110
ro.build.type=user
ro.build.user=build
ro.build.host=i3-ri-14-use1a-b-16-04-fos-197
ro.build.tags=amz-p,release-keys
ro.build.flavor=athena-user
ro.build.system_root_image=true
ro.product.model=AEOAT
ro.product.brand=Amazon
ro.product.name=athena
ro.product.device=athena
# ro.product.cpu.abi and ro.product.cpu.abi2 are obsolete,
# use ro.product.cpu.abilist instead.
ro.product.cpu.abi=armeabi-v7a
ro.product.cpu.abi2=armeabi
ro.product.cpu.abilist=armeabi-v7a,armeabi
ro.product.cpu.abilist32=armeabi-v7a,armeabi
```

Exhibit 11. build.prop: Fire OS 7.5.6.0, package com.amazon.athena.android.os, Cortex-A53 ISA variant.

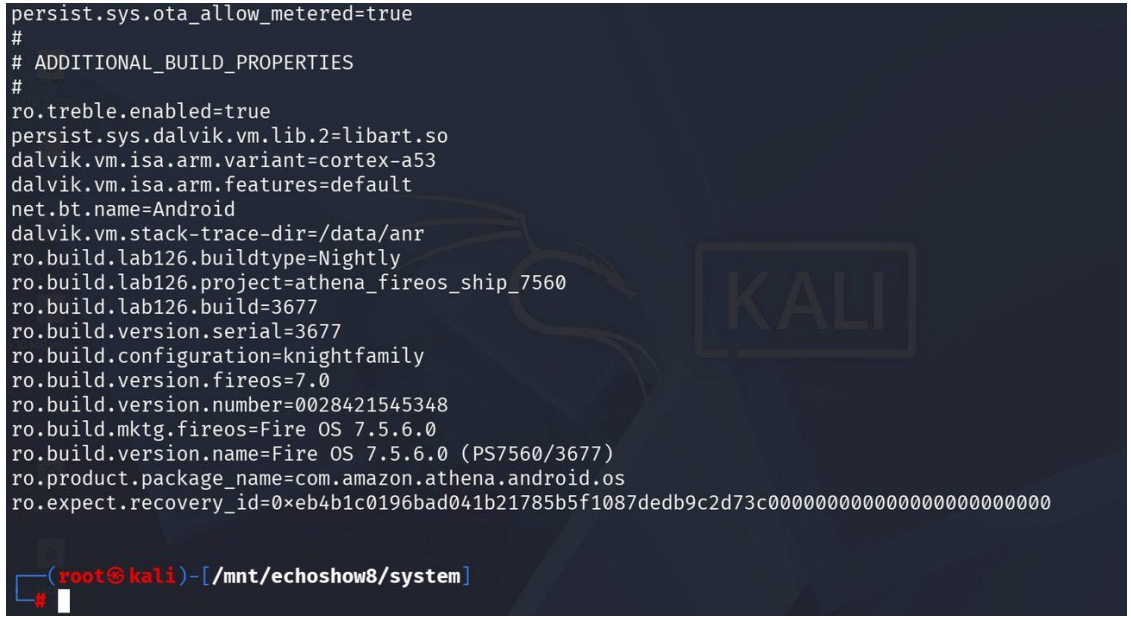


Exhibit 12. /system/app listing (Amazon device, WebView, Bluetooth, KeyChain, Music, etc.).

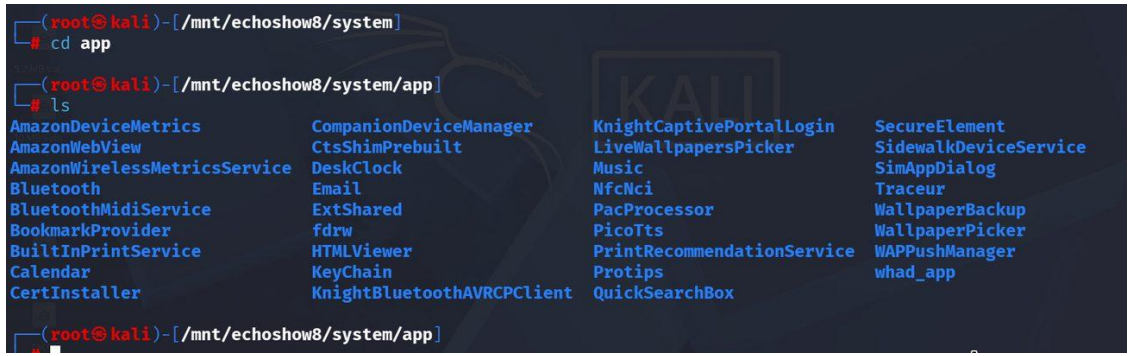


Exhibit 13. /data/data/com.android.settings/shared_prefs listing and deviceNameSharedPref.xml (“Amazing Assistant”).

```
┌──(root㉿kali)-[/mnt]
└─# cd /mnt/echoshow8/data/com.android.settings/shared_prefs

┌──(root㉿kali)-[/mnt/echoshow8/data/com.android.settings/shared_prefs]
└─# ls
account-verification.xml                    com.android.settings.FLAGS.xml
alexa-pairing-mode.xml                      com.android.settings_preferences.xml
bluetooth_settings.xml                      com.android.settings.xml
com.amazon.settings.bluetooth_prefs_key.xml development.xml
com.amazon.settings.CONFIG_BUNDLE_STORE.xml device_disconnection_times.xml
com.amazon.settings.DNSVR_STORE.xml         deviceNameSharedPref.xml
com.amazon.settings.SETTINGS_STORE.xml      KBluetoothControllerSharedPrefs.xml
com.amazon.settings.wifi.xml                KnightSettingsMetrics.xml

┌──(root㉿kali)-[/mnt/echoshow8/data/com.android.settings/shared_prefs]
└─# cat deviceNameSharedPref.xml

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <string name="deviceNameKey">Amazing Assistant</string>
</map>

┌──(root㉿kali)-[/mnt/echoshow8/data/com.android.settings/shared_prefs]
└─#
```

Exhibit 14. /data/system/users/userlist.xml and sorteduserlist.xml – a single user (id 0) and guest restrictions.

```
┌──(kali㉿kali)-[/mnt/echoshow8/system/users]
└─$ cat userlist.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<users nextSerialNumber="10" version="7">
    <guestRestrictions>
        <restrictions no_sms="true" no_install_unknown_sources="true" no_config_wifi="true" no_outgoing_calls="t
rue" />
    </guestRestrictions>
    <deviceOwnerUserId id="-10000" />
    <user id="0" />
</users>

┌──(kali㉿kali)-[/mnt/echoshow8/system/users]
└─$ cat sorteduserlist.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<users>
<user id="0" />
</users>

┌──(kali㉿kali)-[/mnt/echoshow8/system/users]
└─$
```

Exhibit 15. /data/system/users/0.xml – user "Sarfraz", lastLoggedIn 1698016596033, build fingerprint 0028421545216. The date command output (year 55778) is a unit error – see Appendix E.

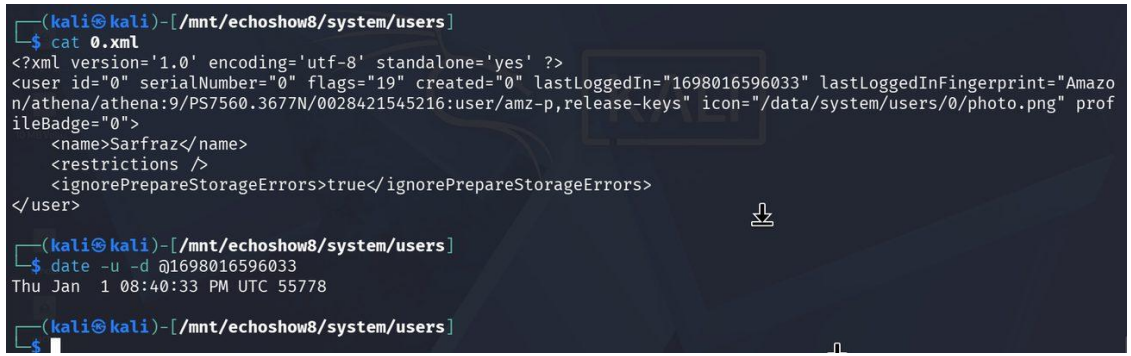


Exhibit 16. /data/system/usagestats/0/daily – nine epoch-millisecond file names; one converted by truncating to seconds.

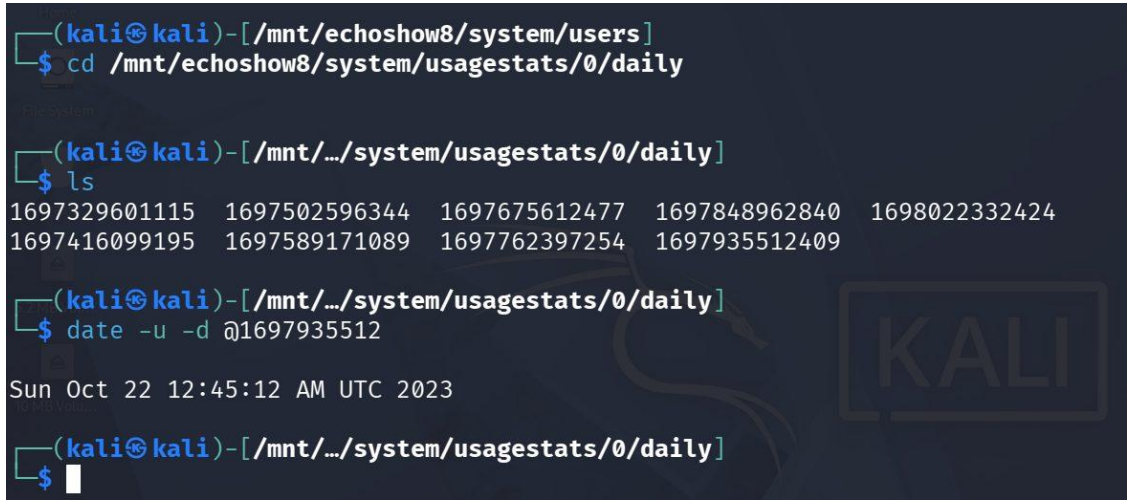


Exhibit 17. usagestats daily file content: interactive/non-interactive counts and per-package lastTimeActive values (many negative, relative).

```
┌──(kali㉿kali)-[/mnt/…/system/usagestats/0/daily]
└─$ cat 1697935512409 | more

<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<usagestats version="1" endTime="85500007">
    <interactive count="11" time="1697860050653" />
    <non-interactive count="11" time="1697859985758" />
    <keyguard-shown count="0" time="0" />
    <keyguard-hidden count="0" time="0" />
    <packages>
        <package lastTimeActive="-1697775240571" package="com.android.providers.calendar" timeActive="0" lastEve
nt="0" />
        <package lastTimeActive="78119740" package="com.amazon.device.rdmapplication" timeActive="0" lastEvent="
0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.whisperjoin.middleware" timeActive="0" last
Event="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.alexa.beaconbroadcaster" timeActive="0" las
tEvent="0" />
        <package lastTimeActive="84396503" package="com.amazon.alexa.datastore.app" timeActive="0" lastEvent="0"
 />
        <package lastTimeActive="59193390" package="com.amazon.alexa.ms.kinex" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.device.smarthome.dshs.services" timeActive=
"0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.imp" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.android.launcher3" timeActive="0" lastEvent="0" />
```

Exhibit 18. /data/system_ce/0/accounts_ce.db – tables and accounts rows (Amazon sync accounts, “Amazon Video Sync”, “Sarfraz”).

```
┌──(kali㉿kali)-[/mnt/…/system/usagestats/0/daily]
└─$ cd /mnt/echoshow8/system_ce/0


┌──(kali㉿kali)-[/mnt/echoshow8/system_ce/0]
└─$ ls
accounts_ce.db  accounts_ce.db-shm  accounts_ce.db-wal  recent_tasks  shortcut_service  snapshots

┌──(kali㉿kali)-[/mnt/echoshow8/system_ce/0]
└─$ sqlite3 accounts_ce.db
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .tables
accounts          android_metadata  authtokens        extras
sqlite> SELECT * FROM accounts;
1|AmazonSyncAccount-cba1f8294651551b7d9d16bef0970fdd79fca92f414c3338137185f6acf40f71|amazon.account|
3|Amazon Video Sync|amazon.account|
4|AmazonSyncAccount-6957b355b974dde07bb47bf8cd86218da71f3ce800784526ba4345cda0abf63b|amazon.account|
5|Sarfraz|com.amazon.account|
sqlite>
```

Exhibit 19. /data/system_de/0/accounts_de.db – accounts rows with creation epochs, and start of the meta table.

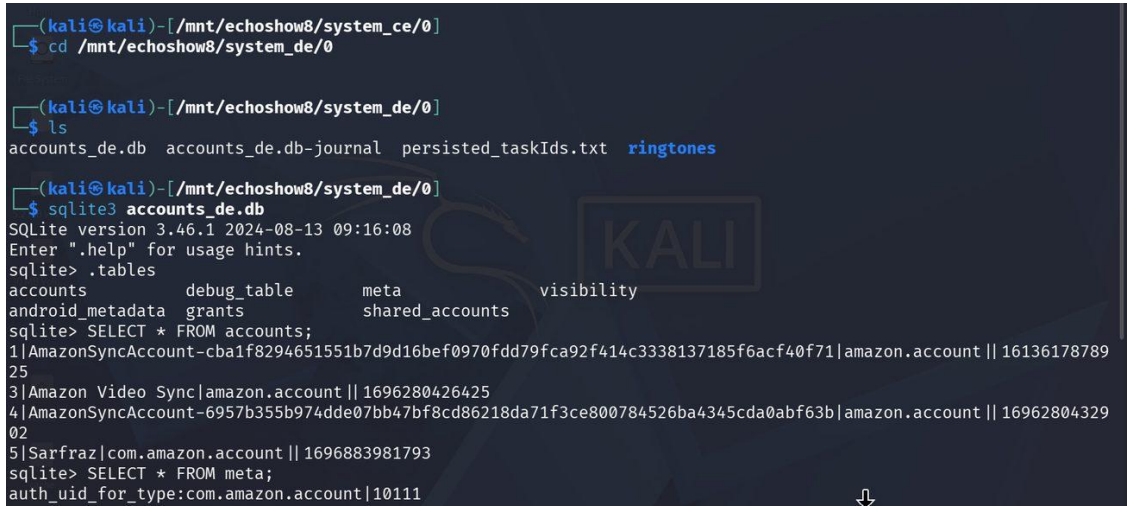


Exhibit 20. accounts_de.db debug_table – account add/remove actions with textual timestamps.

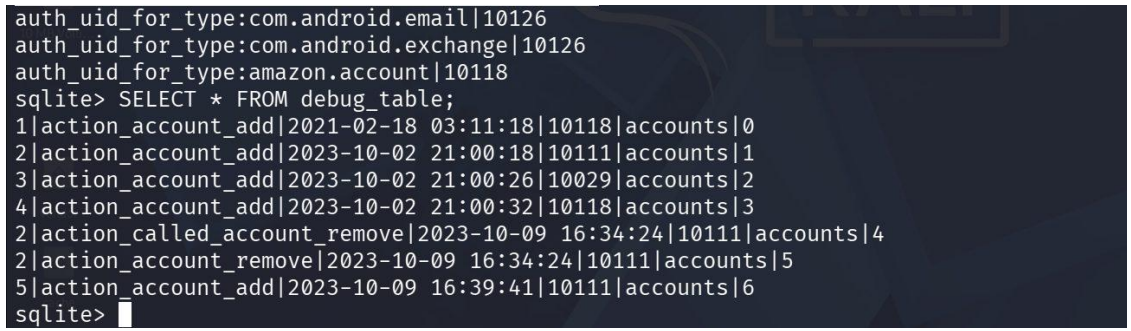


Exhibit 21. /data/system_ce/0/snapshots – five task snapshots (617, 623–625, 627) with .proto and “reduced” copies.

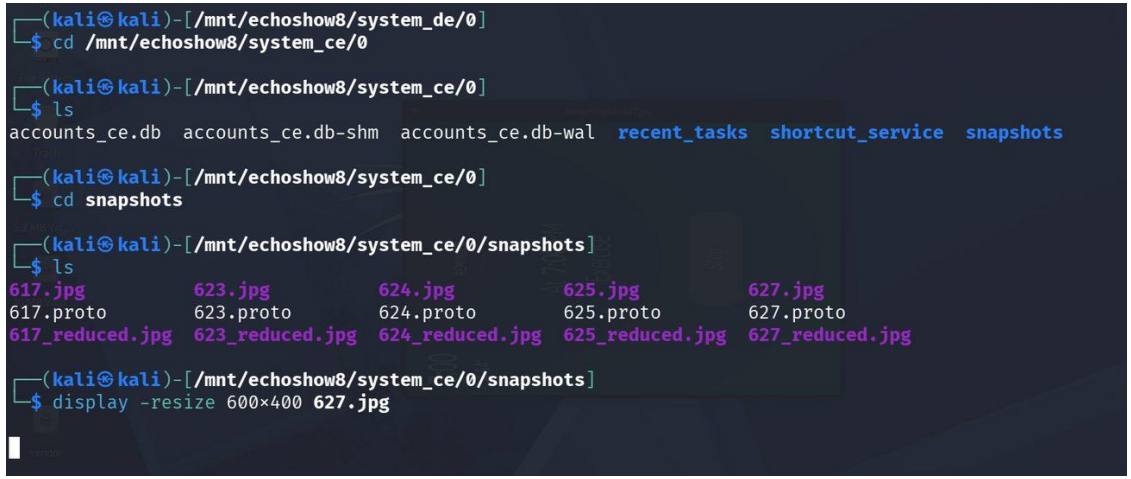


Exhibit 22. Snapshot 617.jpg – YouTube-style video page on the device screen.



Exhibit 23. Snapshot 627.jpg – on-device “Exercise” reminder at 7:00 PM.



Exhibit 24. Snapshot 623.jpg – on-device Bluetooth settings screen naming a paired laptop.

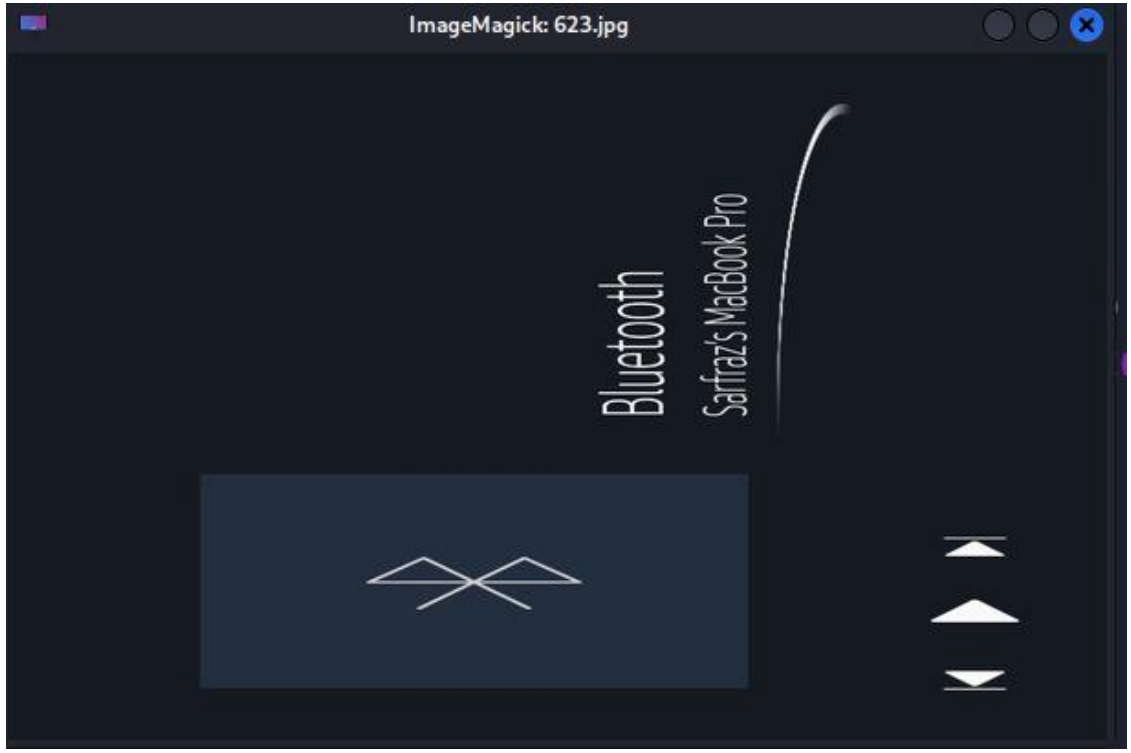


Exhibit 25. /data/system_ce/0/recent_tasks/617_task.xml – intent searching YouTube for “kapil sharma show”; last_time_moved 1698022572750; date -u conversion of the seconds part.

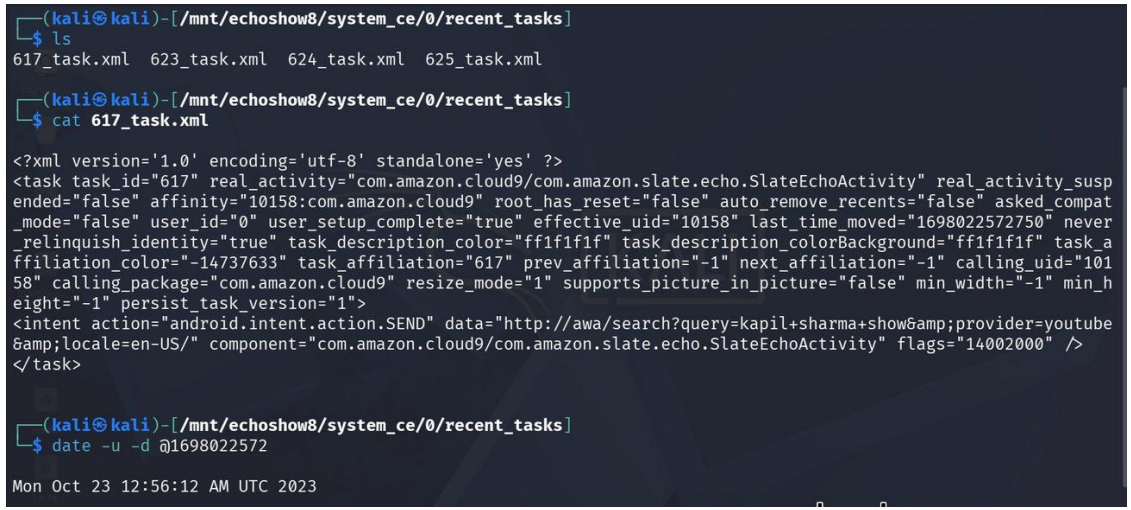


Exhibit 26. bulk_extractor 2.1.1 completion: 7,818 MB processed in 740.3 s; 18,382 email features.

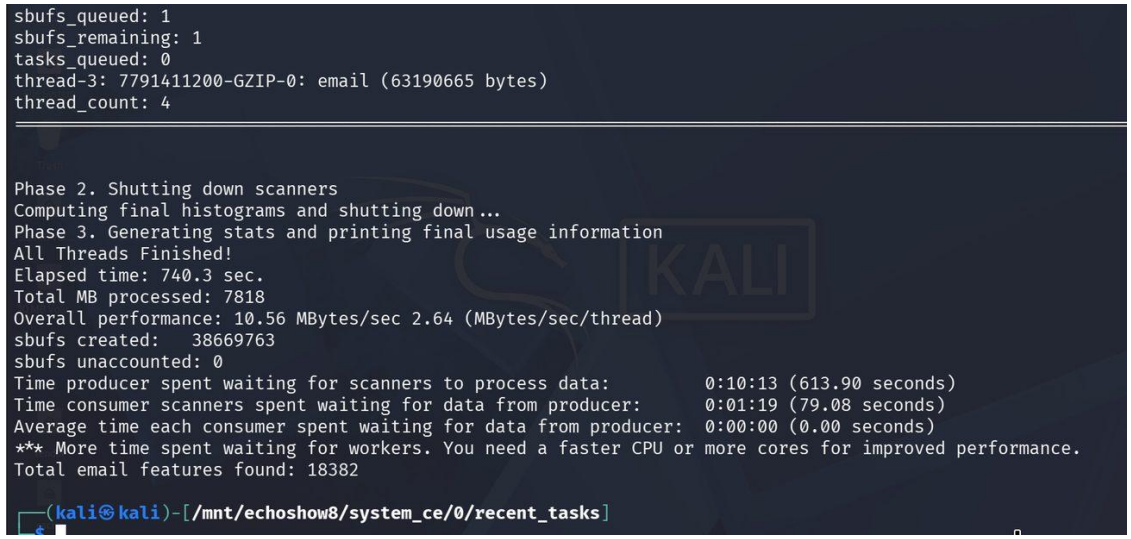


Exhibit 27. bulk_extractor output directory (histograms, url, domain, exif, telephone, vcard and other feature files).

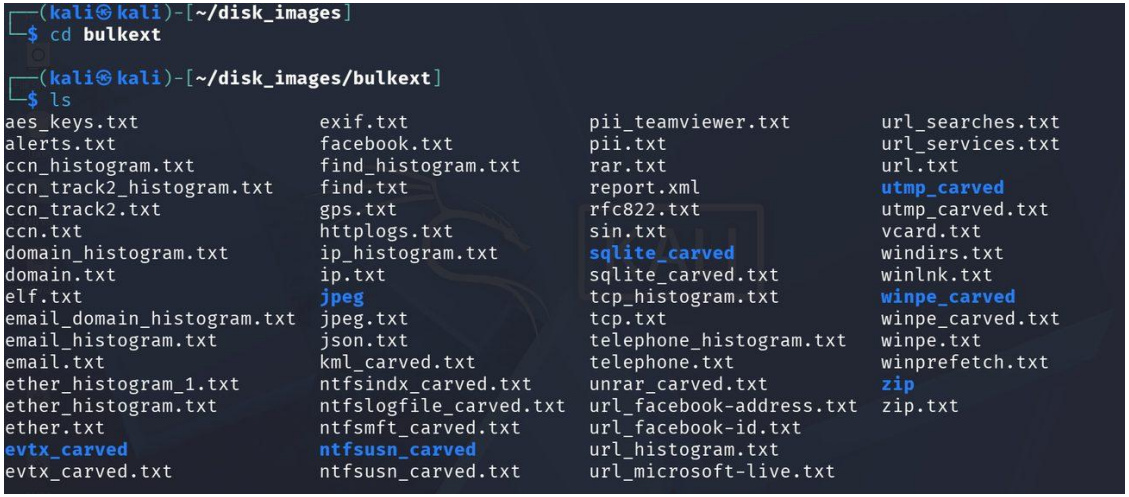


Exhibit 28. /data/data/com.amazon.cloud9/app_amazon_webview/amazon_webview – Chromium-based profile files (History, Cookies, Web Data, Bookmarks, etc.).

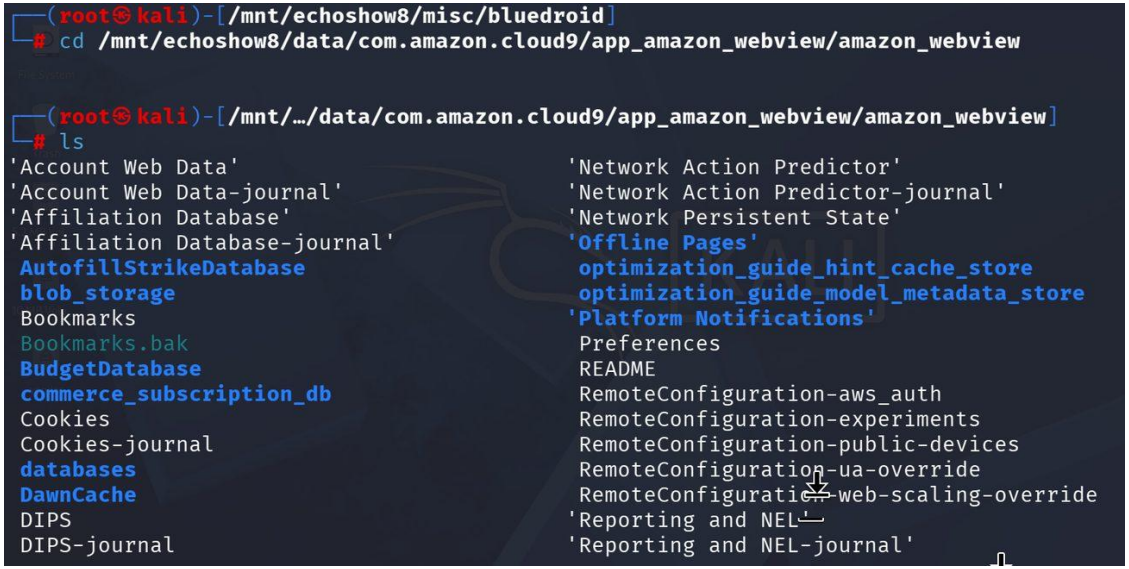


Exhibit 29. History database – visits table (first rows; visit_time in WebKit microseconds).

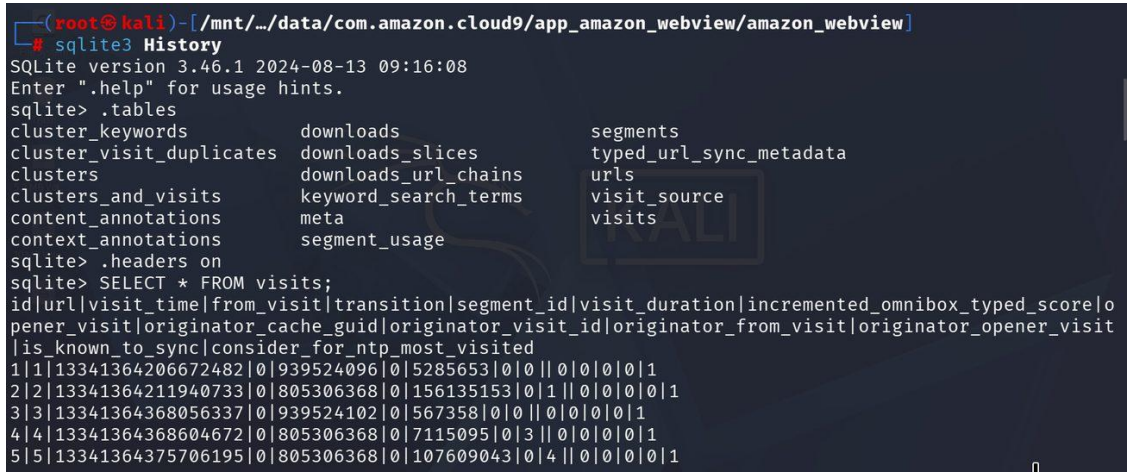


Exhibit 30. History keyword_search_terms – “search chicken nuggets on google” and “close google”.

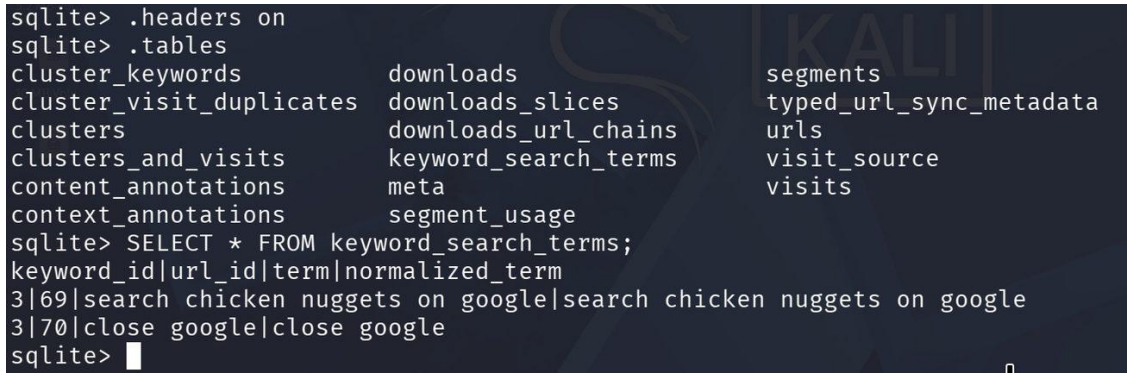
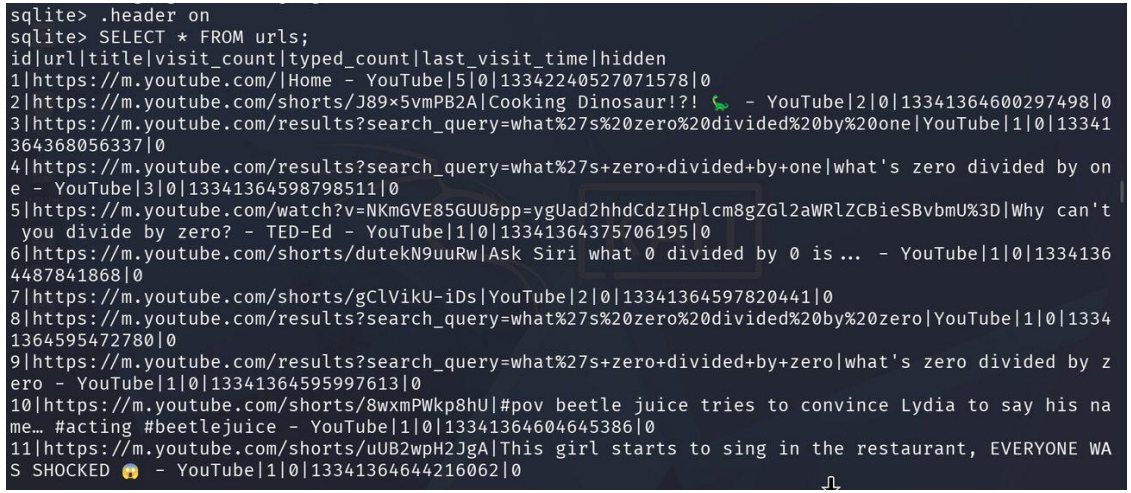
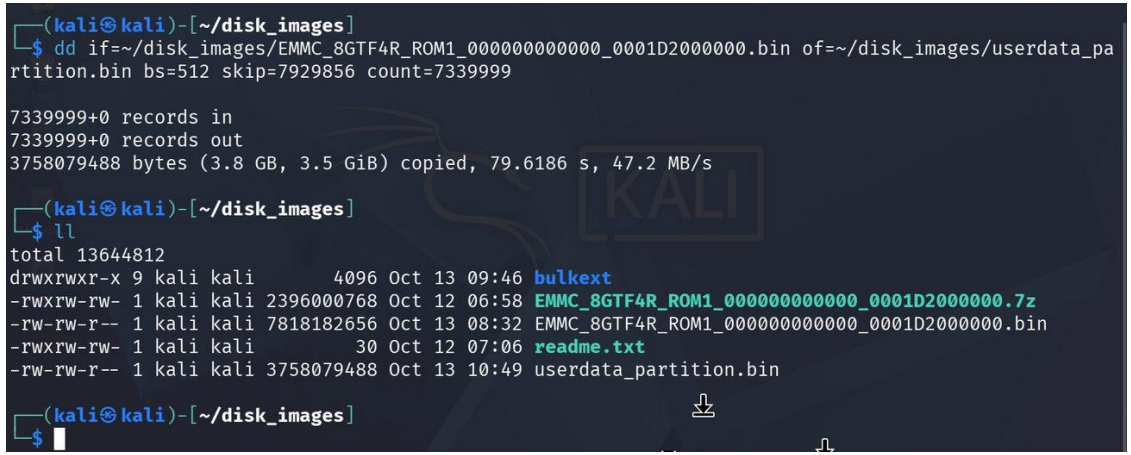


Exhibit 31. History urls table – YouTube search and shorts entries with last_visit_time values.



```
sqlite> .header on
sqlite> SELECT * FROM urls;
id|url|title|visit_count|typed_count|last_visit_time|hidden
1|https://m.youtube.com/|Home - YouTube|5|0|13342240527071578|0
2|https://m.youtube.com/shorts/J89×5vmPB2A|Cooking Dinosaur!?! 🦕 - YouTube|2|0|13341364600297498|0
3|https://m.youtube.com/results?search_query=what%27s%20zero%20divided%20by%20one|YouTube|1|0|13341
364368056337|0
4|https://m.youtube.com/results?search_query=what%27s+zero+divided+by+one|what's zero divided by on
e - YouTube|3|0|13341364598798511|0
5|https://m.youtube.com/watch?v=NKmGVE85GUU&pp=ygUad2hhdCdzIHplcm8gZGl2aWRlZCBieSBvbmU%3D|Why can't
 you divide by zero? - TED-Ed - YouTube|1|0|13341364375706195|0
6|https://m.youtube.com/shorts/dutekN9uuRw|Ask Siri what 0 divided by 0 is… - YouTube|1|0|1334136
4487841868|0
7|https://m.youtube.com/shorts/gClVikU-iDs|YouTube|2|0|13341364597820441|0
8|https://m.youtube.com/results?search_query=what%27s%20zero%20divided%20by%20zero|YouTube|1|0|1334
1364595472780|0
9|https://m.youtube.com/results?search_query=what%27s+zero+divided+by+zero|what's zero divided by z
ero - YouTube|1|0|13341364595997613|0
10|https://m.youtube.com/shorts/8wxmPWkp8hU|#pov beetle juice tries to convince Lydia to say his na
me… #acting #beetlejuice - YouTube|1|0|13341364604645386|0
11|https://m.youtube.com/shorts/uUB2wpH2JgA|This girl starts to sing in the restaurant, EVERYONE WA
S SHOCKED 😱 - YouTube|1|0|13341364644216062|0
```

Exhibit 32. dd of the userdata region (skip=7929856, count=7339999, bs=512) to userdata_partition.bin (3,758,079,488 bytes).



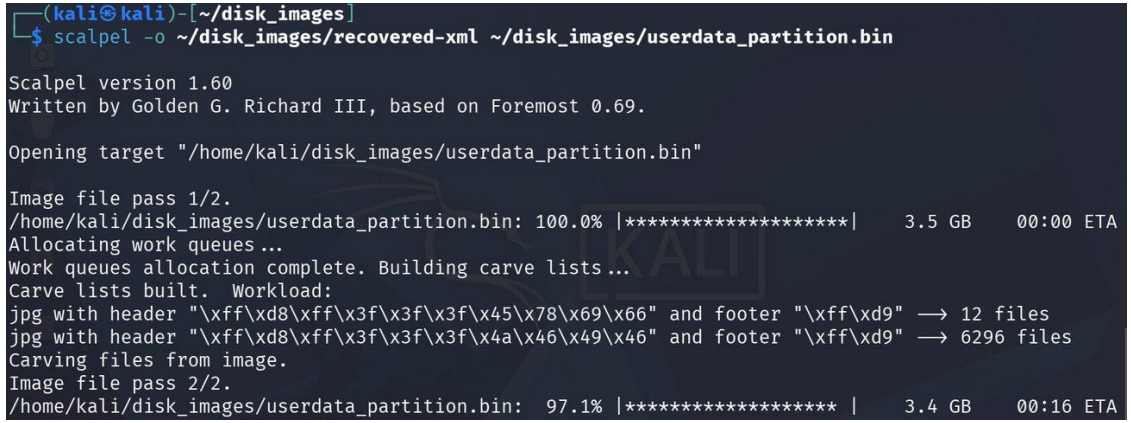
```
┌──(kali㉿kali)-[~/disk_images]
└─$ dd if=~/disk_images/EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin of=~/disk_images/userdata_pa
rtition.bin bs=512 skip=7929856 count=7339999

7339999+0 records in
7339999+0 records out
3758079488 bytes (3.8 GB, 3.5 GiB) copied, 79.6186 s, 47.2 MB/s

┌──(kali㉿kali)-[~/disk_images]
└─$ ll
total 13644812
drwxrwxr-x 9 kali kali       4096 Oct 13 09:46 bulkext
-rwxrw-rw- 1 kali kali 2396000768 Oct 12 06:58 EMMC_8GTF4R_ROM1_000000000000_0001D2000000.7z
-rw-rw-r-- 1 kali kali 7818182656 Oct 13 08:32 EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
-rwxrw-rw- 1 kali kali         30 Oct 12 07:06 readme.txt
-rw-rw-r-- 1 kali kali 3758079488 Oct 13 10:49 userdata_partition.bin

┌──(kali㉿kali)-[~/disk_images]
└─$
```

Exhibit 33. Scalpel 1.60 run on userdata_partition.bin: 12 + 6,296 JPEG candidate headers.



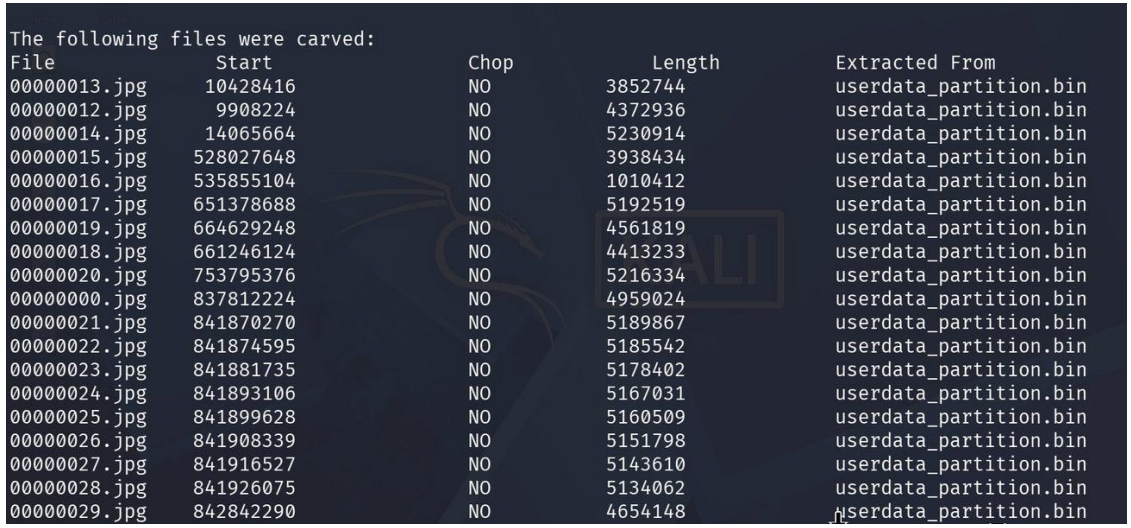
```
┌──(kali㉿kali)-[~/disk_images]
└─$ scalpel -o ~/disk_images/recovered-xml ~/disk_images/userdata_partition.bin

Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/kali/disk_images/userdata_partition.bin"

Image file pass 1/2.
/home/kali/disk_images/userdata_partition.bin: 100.0% |********************|    3.5 GB    00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built.  Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" ⟶ 12 files
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" ⟶ 6296 files
Carving files from image.
Image file pass 2/2.
/home/kali/disk_images/userdata_partition.bin:  97.1% |******************** |    3.4 GB    00:16 ETA
```

Exhibit 34. Scalpel audit.txt – carved file offsets and lengths within userdata_partition.bin.



```
The following files were carved:
File              Start              Chop          Length          Extracted From
00000013.jpg     10428416            NO          3852744          userdata_partition.bin
00000012.jpg      9908224            NO          4372936          userdata_partition.bin
00000014.jpg     14065664            NO          5230914          userdata_partition.bin
00000015.jpg    528027648            NO          3938434          userdata_partition.bin
00000016.jpg    535855104            NO          1010412          userdata_partition.bin
00000017.jpg    651378688            NO          5192519          userdata_partition.bin
00000019.jpg    664629248            NO          4561819          userdata_partition.bin
00000018.jpg    661246124            NO          4413233          userdata_partition.bin
00000020.jpg    753795376            NO          5216334          userdata_partition.bin
00000000.jpg    837812224            NO          4959024          userdata_partition.bin
00000021.jpg    841870270            NO          5189867          userdata_partition.bin
00000022.jpg    841874595            NO          5185542          userdata_partition.bin
00000023.jpg    841881735            NO          5178402          userdata_partition.bin
00000024.jpg    841893106            NO          5167031          userdata_partition.bin
00000025.jpg    841899628            NO          5160509          userdata_partition.bin
00000026.jpg    841908339            NO          5151798          userdata_partition.bin
00000027.jpg    841916527            NO          5143610          userdata_partition.bin
00000028.jpg    841926075            NO          5134062          userdata_partition.bin
00000029.jpg    842842290            NO          4654148          userdata_partition.bin
```

Exhibit 35. df -h – /dev/loop0p23 (3.4 G, 68 % used) mounted at /mnt/echoshow8.

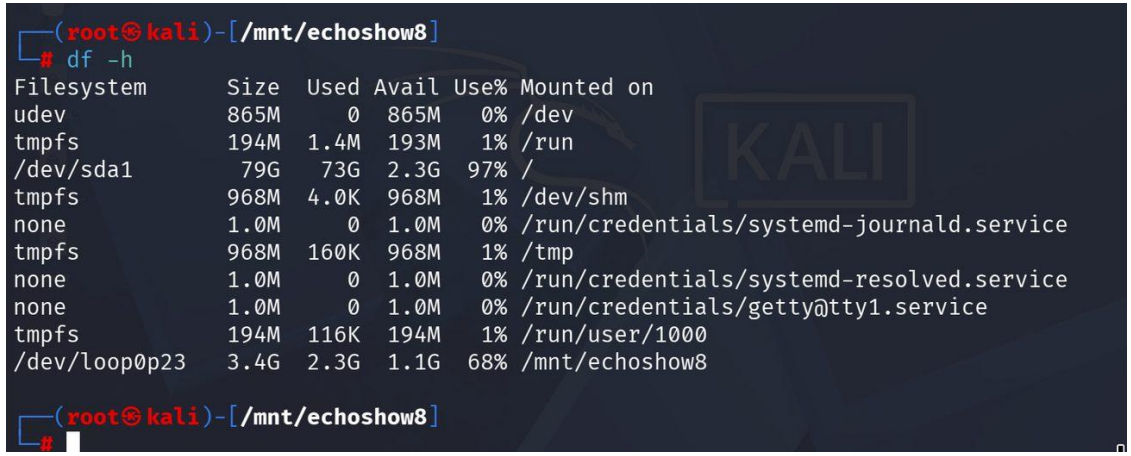


Exhibit 36. foremost 1.5.7 JPEG carve of the full eMMC image.

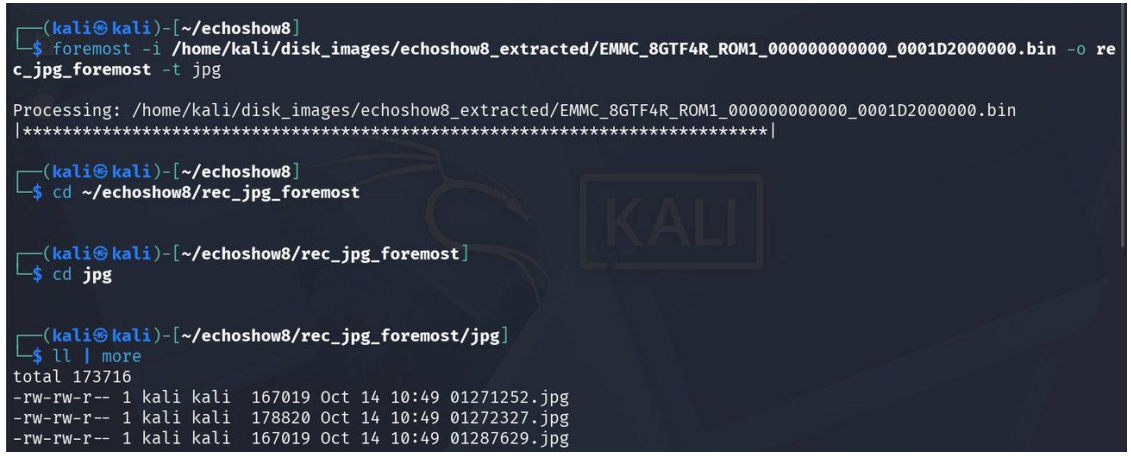


Exhibit 37. foremost result: 6,390 files extracted (jpg = 6390).

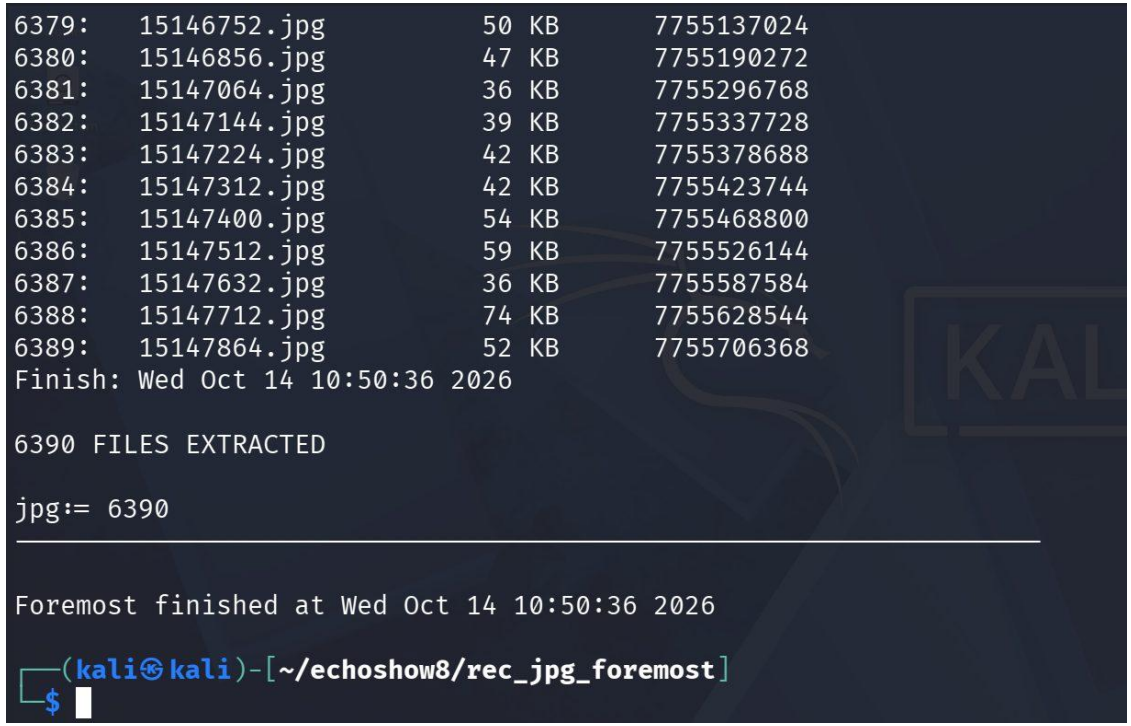


Exhibit 38. Thumbnail view of carved JPEGs – predominantly cached streaming and music artwork.

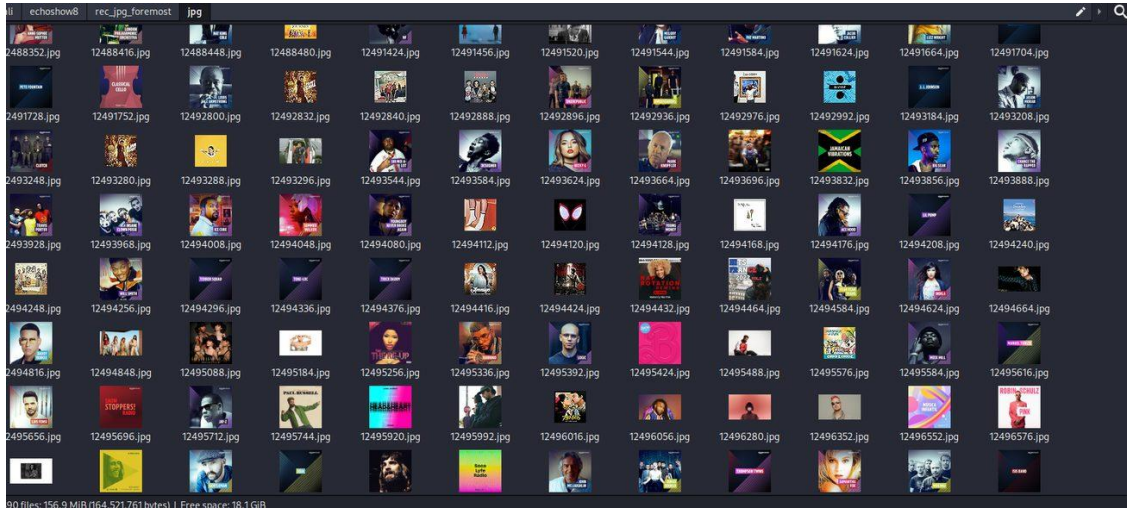


Exhibit 39. exiftool on a carved JPEG: 480×480 JPEG; File Modification/Access times are the examiner's extraction times, not original evidence.

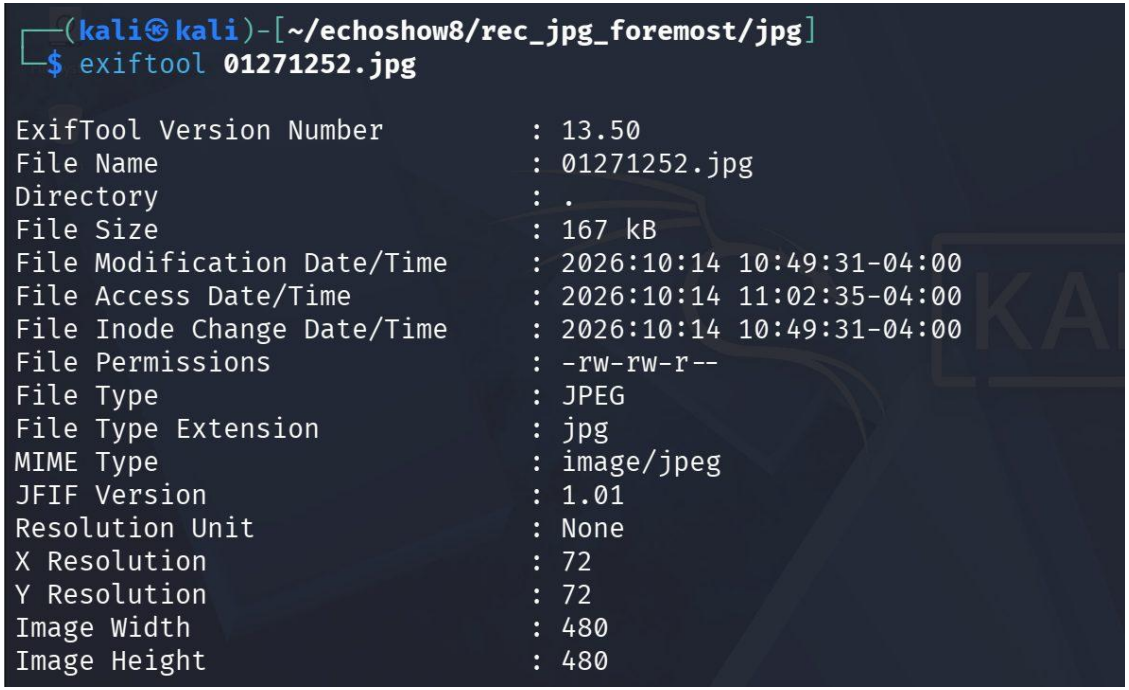


Exhibit 40. /data/data/com.amazon.avod/files/databases – Prime Video databases (dbplaybackhistory, bookmark, event, mediaEvents).

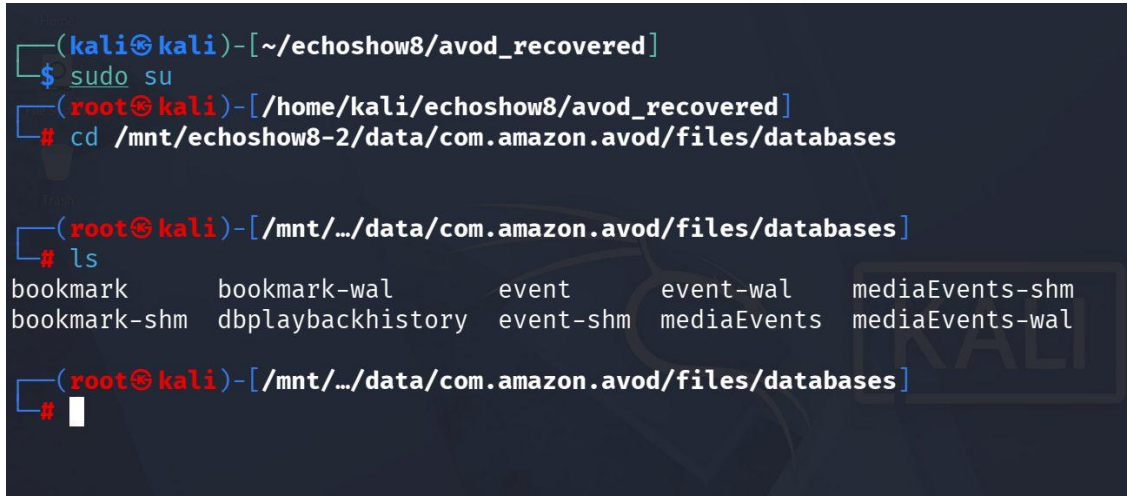


Exhibit 41. dbplaybackhistory (working copy in /tmp) – playbackHistory table header and first records (cropped to omit identifiers).



Exhibit 42. com.amazon.zordon mixtape.db upload_requests (rows 1–5): photo_booth files, sizes, states and MD5 values.

```
sqlite> .headers on
sqlite> SELECT * FROM cloud_media_parents;
sqlite> .headers on
sqlite> SELECT * FROM upload_requests;
_id|queue|parent_id|file_uri|file_size|name|priority|state|suppress_deduplication|rename_on_name_conflict|retry
_exceptions|last_error|attempts|md5|local_id|node_id|bytes_sent|bytes_received|category|cds_path|app_data|conte
nt_date|add_to_family_archive|created_date|visual_digest
1|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_single_shot2023-10
-02_17-52-26-169.jpeg|2849322|photo_booth_single_shot2023-10-02_17-52-26-169.jpeg|0|DONE|0|0|0||1|05fec6d31b5ed
3fae922fd92da90e3a1||AydH_yW8SVWt9Ei-YIWhTg|2849322|2849322|jpeg|||0||
2|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_single_shot2023-10
-07_13-25-13-104.jpeg|3010907|photo_booth_single_shot2023-10-07_13-25-13-104.jpeg|0|DONE|0|0|0||1|22907c18e2bc9
b38b19892cf3642d77c||MZSCkr7aSNGJneyNOGyV_Q|3010907|3010907|jpeg|||0|395|
3|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_single_shot2023-10
-09_15-34-15-361.jpeg|3353396|photo_booth_single_shot2023-10-09_15-34-15-361.jpeg|0|DONE|0|0|0||1|d6a484656962a
ea314ae90cb7286ce14||AMuRAtPvTciCglQUfDBbMg|3353396|3353396|jpeg|||0|395|
4|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_video1696880081230
.mp4|6047662|photo_booth_video1696880081230.mp4|0|DONE|0|0|0||1|79c220a31975e66d377e12e5b182b062||jtNsNiZBT5GWo
mjI5cxg7A|6047662|6047662|mp4|||0|395|
5|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_single_shot2023-10
-09_19-45-07-628.jpeg|2935516|photo_booth_single_shot2023-10-09_19-45-07-628.jpeg|0|DONE|0|0|0||1|4c675e3d566f2
df6b9f298acaab92c1d||fYRDY3pOTcmqyRfE9oUUBQ|2935516|2935516|jpeg|||0|395|
```

Exhibit 43. upload_requests rows 6–8; row 8 (video, 13,958,448 bytes) is in state QUEUED.

```
6|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_single_shot2023-10
-17_20-29-28-381.jpeg|2717919|photo_booth_single_shot2023-10-17_20-29-28-381.jpeg|0|DONE|0|0|0||1|6823f8c4bb1b8
05f873a28a592a78beb||rr1nGSfQRnCXp_egEgIsMg|2717919|2717919|jpeg|||0|395|
7|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_single_shot2023-10
-21_19-02-39-687.jpeg|2961976|photo_booth_single_shot2023-10-21_19-02-39-687.jpeg|0|DONE|0|0|0||1|f2acb3eecc567
831c91e33682d2c6e06||wqNbnDllRnyGnmk_Q7eDkQ|2961976|2961976|jpeg|||0|395|
8|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_video1697929429134
.mp4|13958448|photo_booth_video1697929429134.mp4|0|QUEUED|0|0|0||0|||0|0|||||0|395|
sqlite>
```

Exhibit 44. cloud_nodes row 7 (photo created 1697929372480; image date/time 1697914959000; model AEOAT).

```
7|7|wqNbnDllRnyGnmk_Q7eDkQ||1697929372480|||||1|1697929373700|photo_booth_single_shot2023-10-21_19-02-39-687.jp
eg||0|1|2|image/jpeg|jpg|f2acb3eecc567831c91e33682d2c6e06|1f25eba55c6473c94c2cbaa0f417f7ef|1697929359067||169791
4959067|2961976|1|||||sRGB|1697914959000|1697914959000|1697914959000|Auto|Not Defined|1/24|No Flash|3.8||2160||
|Amazon|Center-weighted average|AEOAT|1|Pixels/Inch|||custom software|Auto|3456|||1|0|1|0|||||||||||||||||||0|
0|5|0|—1
sqlite>
```

Exhibit 45. date -u conversion of 1697929372 and 1697929464 (Sat Oct 21 2023, 23:02:52 and 23:04:24 UTC).

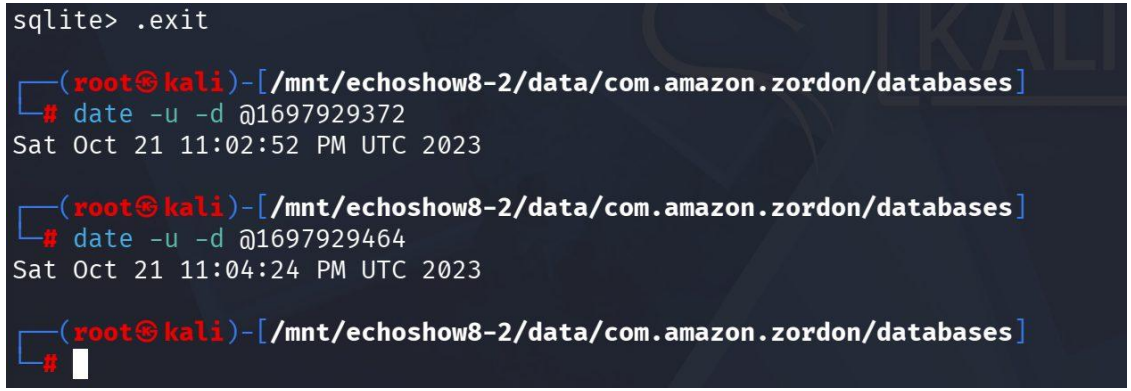


Exhibit 46. foremost carve for mp4 – 0 files extracted.

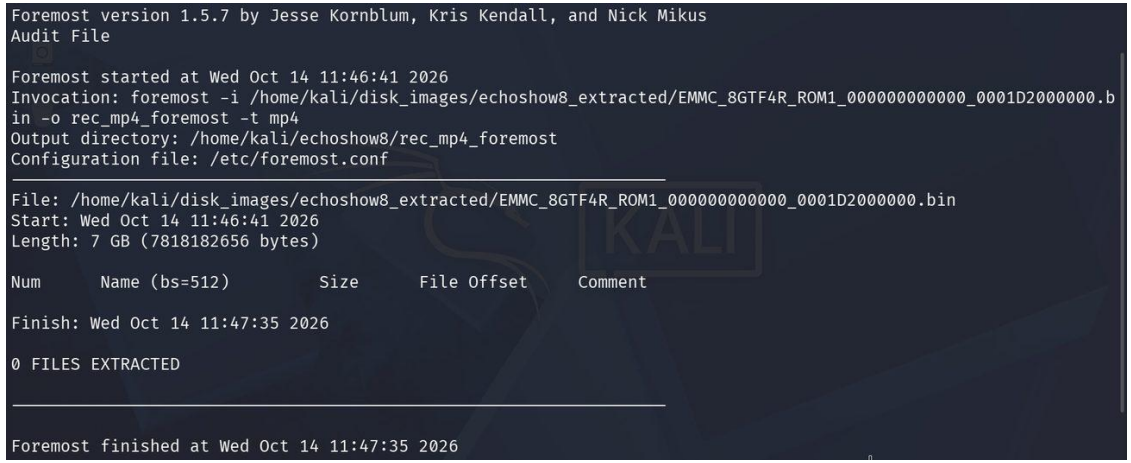


Exhibit 47. alexa-anchor output converted with jq and searched for “money” – Amazon Music SONG impression.

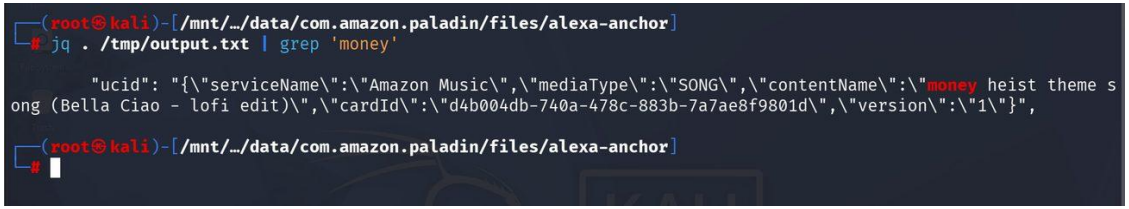


Exhibit 48. message-timestamps file – ISO-8601 timestamps with −0400 offset (2023-10-23T18:28…).

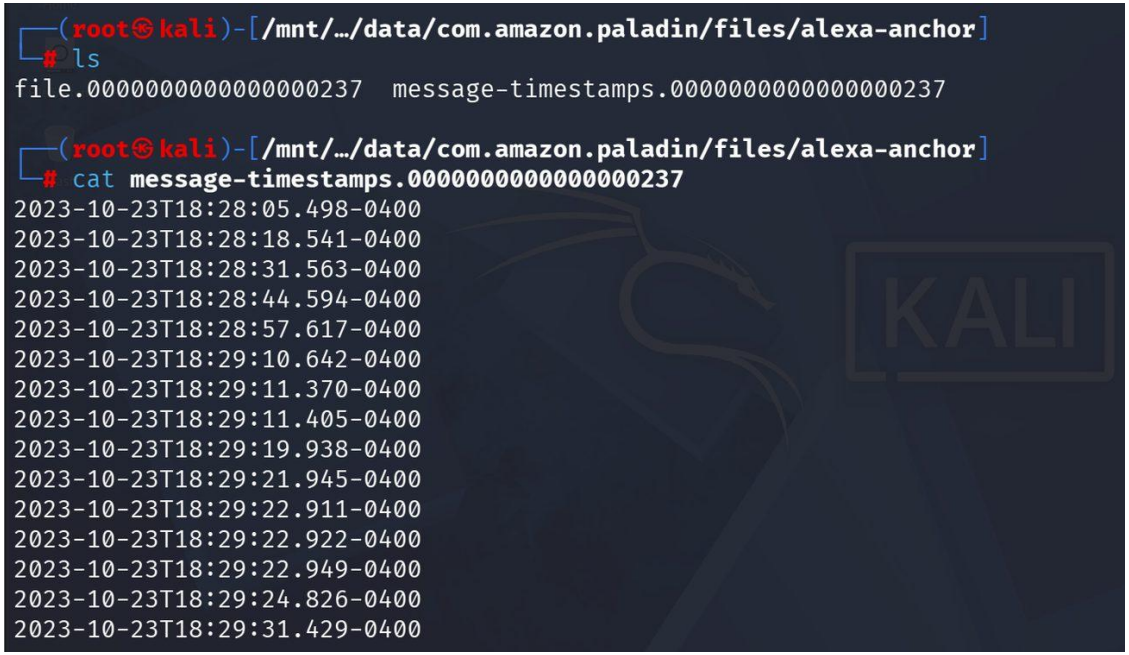


Exhibit 49. PhotoRec 7.2 output (recup_dir.1) – carved ELF-labelled fragments.

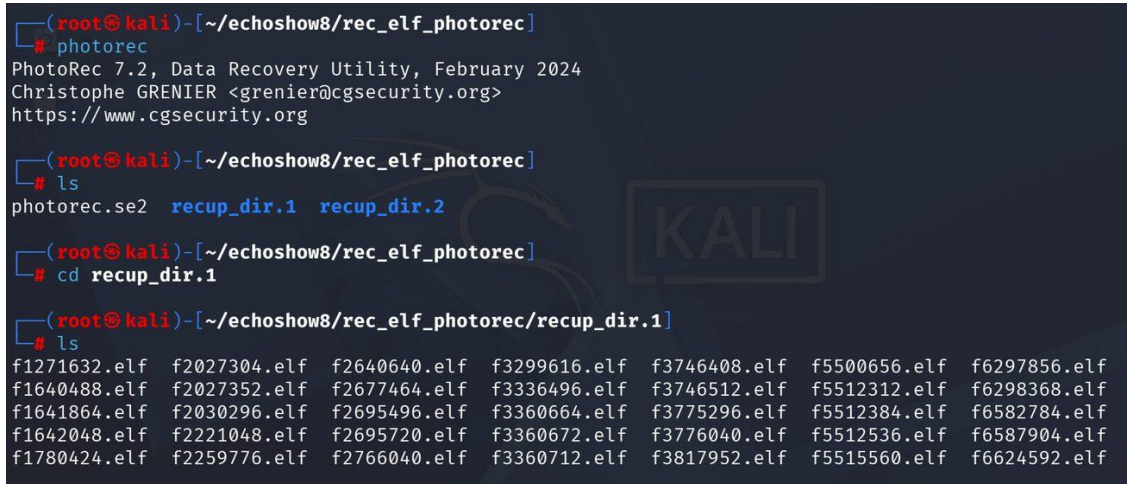


Exhibit 50. grep “money” across the PhotoRec output – 14 binary matches.

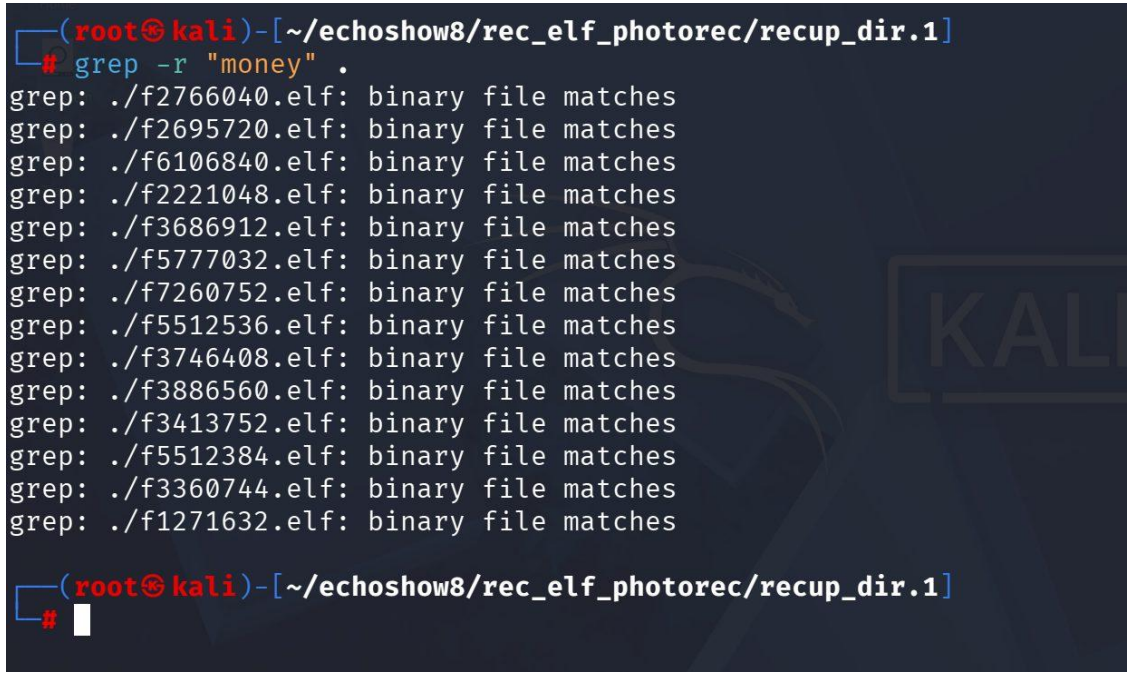


Exhibit 51. strings on carved f5777032.elf – alexa-anchor impressions dated 2023-10-22T22:12:50−0400 including the Amazon Music “money heist” SONG.


```
\"camera\":\"ON\"}","impressionId":"ed898d39-7e5e-448a-b27e-de42c1287d72","eventType":"IMPRESSION_END","s
lotPosition":27,"duration":12892,"dismissalReason":"AUTO","ucid":"2cd69232-8f01-415c-a427-78940ba4a527","
attentionSlot":56}},{"data":{"schemaId":"alexa_anchor.DeviceImpression.3","timestamp":"2023-10-22T22:12:5
0.917-0400","producerId":"alexa-anchor","messageId":"aa40a70f-3c6c-42dd-99f0-1e0ed7de8ca6","sessionId":"8
b9caeac-da7c-4e2c-a9ed-3f08487e5655","compositionId":"f57d5548-f8a8-45ed-a2a8-7b3500ee7692","cardName":"R
ecentlyPlayedAudioContentProvider","itemMeta":"{\"IsHestia\":true,\"isPersonalized\":false,\"cardType\":\
"\"}","placementId":"a4ae9aca-768a-4f81-946a-be86e6c1f1f7","personId":"UNDEF","deviceState":"{\"cohort\":
\"UNKNOWN\",\"deviceOpMode\":\"DEFAULT\",\"privacyMode\":\"OFF\",\"cameraShutter\":\"OFF\",\"camera\":\"O
N\"}","impressionId":"8b1f7dce-11ce-4249-883f-ba96fd6c7cef","eventType":"IMPRESSION_END","slotPosition":2
8,"duration":12862,"dismissalReason":"AUTO","ucid":"{\"serviceName\":\"Amazon Music\",\"mediaType\":\"SON
G\",\"contentName\":\"money heist theme song (Bella Ciao - lofi edit)\",\"cardId\":\"274640fc-fe76-4ffb-a
228-8c432f6813e1\",\"version\":\"1\"}","attentionSlot":57}},{"data":{"schemaId":"alexa_anchor.DeviceImpre
ssion.3","timestamp":"2023-10-22T22:13:03.942-0400","producerId":"alexa-anchor","messageId":"94c1e288-7a9
4-4a7c-b360-9679e5bcba1f","sessionId":"8b9caeac-da7c-4e2c-a9ed-3f08487e5655","compositionId":"f57d5548-f8
a8-45ed-a2a8-7b3500ee7692","cardName":"PhotoCard-AmazonPhotos","itemMeta":"{\"IsHestia\":true,\"isPersona
lized\":false,\"cardType\":\"\"}","placementId":"760cf067-8bf2-485c-931a-e4192459adec","personId":"UNDEF"
,"deviceState":"{\"cohort\":\"UNKNOWN\",\"deviceOpMode\":\"DEFAULT\",\"privacyMode\":\"OFF\",\"cameraShut
ter\":\"OFF\",\"camera\":\"ON\"}","impressionId":"cd7ccf7c-925d-451f-8406-80cf43577716","eventType":"IMPR
ESSION_END","slotPosition":29,"duration":12878,"dismissalReason":"AUTO","ucid":"Rbo2iIr_QWy_u5PNS9ajew","
attentionSlot":58}},{"data":{"schemaId":"alexa_anchor.DeviceImpression.3","timestamp":"2023-10-22T22:13:1
6.977-0400","producerId":"alexa-anchor","messageId":"66ed4fdf-aa0e-4faf-89a7-bf065e745f28","sessionId":"8
```


Appendix C – Commands and queries

Command / query	Purpose
7z x EMMC_8GTF4R_ROM1_000000000000_0001D2000000.7z	Extract archive
md5sum EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin	Hash extracted image
sudo losetup --partscan --find --show --read-only /home/kali/disk_images/EMMC…bin	Read-only attach (loop1)

Command / query	Purpose
losetup -	List loop devices (shows loop0 offset 4060086272, sizelimit 3758079488)
lsblk /dev/loop1	Partition table via kernel
mmls EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin	Sleuth Kit partition layout
sudo mkdir -p /mnt/echoshow8; sudo losetup -fP ~/disk_images/EMMC…bin	Second attachment (no read-only flag)
sudo mount -o ro /dev/loop0p21 /mnt/echoshow8	Mount system partition read-only
cat build.prop; cat deviceNameSharedPref.xml; cat userlist.xml; cat 0.xml	Device and user records
date -u -d @1697935512; date -u -d @1698022572; date -u -d @1697929372	Epoch seconds to UTC
sqlite3 accounts_ce.db / accounts_de.db / History / dbplaybackhistory / mixtape.db / map_data_storage_v2.db	Database queries (SELECT * FROM <table>)
bulk_extractor -o ~/disk_images/bulkext EMMC…bin	Feature extraction (740.3 s)
dd if=EMMC…bin of=userdata_partition.bin bs=512 skip=7929856 count=7339999	Carve userdata partition by sector offset

Command / query	Purpose
scalpel -o ~/disk_images/recovered-xml userdata_partition.bin	JPEG carving (Scalpel 1.60)
foremost -i …EMMC…bin -o rec_jpg_foremost -t jpg; … -t mp4	JPEG and MP4 carving
exiftool 01271252.jpg	Metadata of a carved image
photorec (output rec_elf_photorec); grep -r "money" .; strings f5777032.elf | grep money	Fragment carve and string search
cat file.0000000000000000237 > /tmp/output.txt; jq . /tmp/output.txt	Parse alexa-anchor log

Appendix D – Source paths

Partition	Path
system (loop0p21)	/system/build.prop; /system/app/; /init.recovery.mt8183.rc
userdata (loop0p23)	/data/com.android.settings/shared_prefs/ (deviceNameSharedPref.xml, DNSVR_STORE.xml, wifi.xml)
userdata	/system/users/{0.xml,userlist.xml,sorteduserlist.xml}; /system/usagestats/0/daily/

Partition	Path
userdata	/system_ce/0/{accounts_ce.db,recent_tasks/,snapshots/}; /system_de/0/accounts_de.db
userdata	/misc/wifi/WifiConfigStore.xml; /misc/bluedroid/bt_config.conf; /user_de/0/com.android.bluetooth/shared_prefs/
userdata	/data/com.amazon.cloud9/app_amazon_webview/amazon_webview/History
userdata	/data/com.amazon.avod/files/databases/dbplaybackhistory; …/responseCache/…/Profiles/
userdata	/data/com.amazon.zordon/databases/amzn1.account.<masked>.mixtape.db
userdata	/data/com.amazon.paladin/files/alexa-anchor/{file.…237,message-timestamps.…237}
userdata	/data/com.amazon.kindle.otter.oobe/cache/volley/ (setup cache); /data/amazon.speech.sim/files/

Appendix E – Manual validation

#	Item	Tool output / raw	Manual check	Result
V1	User lastLoggedIn	Tool output: date -u -d @1698016596033 printed “Thu Jan 1 08:40:33 PM UTC 55778”.	Value is in milliseconds. Dividing by 1,000 gives 1698016596 = 2023-10-22 23:16:36 UTC (19:16:36 EDT).	Differed (examiner input error), then matched after correction.
V2	Photo timestamps (cloud_nodes vs file name vs EXIF-style)	Node created 1697929372480; file name 2023-10-21_19-02-39-687; image time 1697914959000.	Node → 23:02:52 UTC; image time read as UTC → 19:02:39 (local); file name 19:02:39. Offset exactly 4 h; 13-second gap to node creation is upload latency.	Matched (EDT).
V3	Recent task 617	date -u -d @1698022572 → Mon Oct 23 00:56:12 UTC 2023	Truncating the millisecond value 1698022572750 to seconds is correct; the .750 fraction is dropped.	Matched.
V4	Build date	ro.build.date.utc=1695264110 vs ro.build.date “Thu Sep 21 02:41:50 UTC 2023”	Independent conversion gives 2023-09-21 02:41:50 UTC.	Matched.

#	Item	Tool output / raw	Manual check	Result
V5	Chromium time	History last_visit_time 13342492250476128	(value − 11,644,473,600,000,000) ÷ 1,000,000 = Unix seconds → 2023-10-22 23:50:50 UTC.	Matched by independent calculation; the parser output itself was not shown.
V6	Carve counts (parser comparison)	Scalpel workload 12 + 6,296 header matches vs foremost 6,390 files	Counts differ; the tools use different JPEG signatures; PhotoRec and foremost outputs were not hash-compared.	Differed – explained, not fully verified.

Appendix F – Academic integrity declaration

I confirm that this report reflects my own examination of the authorised course evidence and that I have identified all sources, tools, and assistance used.

Signed Fuseini Imoru Kantuogaa Registration number: C11/26/DFIT/17291 Date: 15 – 10 - 2026